Komorebi Quadratic-Form Engine

Memory-Bounded Structured Solvers with Accuracy-Certified Extreme Tails

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Abstract

Gaussian quadratic forms occur throughout statistics, numerical analysis, signal processing, kernel methods, and variance-component inference. This work develops a single computational framework for central positive, central indefinite, and positive noncentral forms, with particular emphasis on two difficult regimes: operators too large to reside in accelerator memory and survival probabilities many hundreds or thousands of decades into the tail.

The method separates storage, structural propagation, rank compression, and spectral tail evaluation. Exact block identities permit out-of-core factor actions without changing the represented operator. Polyphase light-cone theorems and certified approximate light cones turn structural locality into phase-aware scheduling. Range-residual invariants quantify unresolved spectral mass and trigger exact residual completion when the remaining rank is small. The final tail layer uses exponential tilting, query-adaptive spectral packets, log-domain accumulation, and explicit support and Fourier-mode remainder bounds.

For exact supplied positive spectra, a directed-rounding backend returns a logarithmic probability interval and an accuracy-driven controller refines internal resolution until the requested multiplicative probability accuracy is proved or fails closed. On the exact control $0.02\chi^2_{80}$, the solver certifies tails through 10^{-5000}; at the 10^{-5000} point a 10^{-18} requested relative accuracy is met with a certified ratio-minus-one of approximately 9.2×10^{-20}. Additional validation includes a rank-1997 production-scale positive spectrum near 10^{-922}, a corrected strong-dependence spectrum near 10^{-458}, signed finite-endpoint controls through 10^{-200}, a rank-340 mixed-sign tail near 10^{-82}, and noncentral exact controls through 10^{-40}.

The central design rule is that approximation is an internal computational choice, not the accuracy contract. A result is called certified only when its explicit enclosure or analytic error ledger proves the requested accuracy.

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1 Results at a glance

The engine is organized around a simple contract: preserve the quadratic form while changing its representation, and report a tail probability only at an accuracy that has been explicitly established.

Component	Resolved statement
Central positive form	$X^\top MX$, with $X \sim N(0,\Sigma)$ and $M \succeq 0$, reduces exactly to $\sum_j \lambda_j Z_j^2$.
Central indefinite form	For symmetric M, the same reduction holds with signed λ_j, yielding a signed generalized chi-square form.
Positive noncentral form	$X \sim N(\mu,\Sigma)$, $M \succeq 0$, reduces exactly to a deterministic shift plus $\sum_j \lambda_j \chi_1^2(\delta_j)$.
Memory-bounded factorization	At fixed resident column width c, nuisance-space rank p, and Krylov width s, streamed sample-space actions use $O(nc+ns+cs+np)$ working storage rather than $O(nm)$.
Structural scheduling	Exact and approximate phase light cones certify which operator blocks can affect a finite-depth Krylov query.
Spectral recovery	Range residual trace, square-trace, and spectral-cap information bounds unresolved spectrum; when the remaining rank is small, streamed residual completion restores the exact nonzero spectrum.
Extreme-tail contract	For exact positive spectra, the public input is a requested multiplicative accuracy ε. The solver returns $[p_-, p_+]$ only as certified when $p_+/p_- - 1 \le \varepsilon$.
Deepest rigorous control	For $0.02\chi^2_{80}$ at $p = 10^{-5000}$, a requested 10^{-18} relative accuracy is certified with achieved ratio-minus-one approximately 9.2×10^{-20}.

The remainder of the manuscript develops these statements from the operator level down to the final probability enclosure.

2 General quadratic-form core

The primary mathematical object is a Gaussian quadratic form.

2.1 Central positive forms

Let

$$X \sim N(0,\Sigma), \qquad M \succeq 0,$$

and write $S = \Sigma^{1/2}$. With $X = SZ$, $Z \sim N(0,I)$,

$$Q = X^\top MX = Z^\top (SMS) Z.$$

If the nonzero eigenvalues of the positive semidefinite operator

$$H = SMS$$

are $\lambda_1, \dots, \lambda_r$, then

$$\boxed{Q \overset{d}{=} \sum_{j=1}^{r} \lambda_j Z_j^2.}$$

The numerical tail problem therefore depends only on a positive spectrum, regardless of how that spectrum was generated.

Proposition 2.1 (Representation equivalence)**.** *The same central quadratic-form law may be supplied by any of the following:*

1. *a nonnegative spectrum* $\{\lambda_j\}$*;*
2. *a positive semidefinite matrix* H*;*
3. *a factor* A *with* $H = AA^\top$ *or* $H = A^\top A$*;*
4. *a replayable streamed factor*
$$A = [A_1 \ \cdots \ A_B].$$

The nonzero spectra of $AA^\top$ *and* $A^\top A$ *are identical, and*

$$AA^\top = \sum_b A_b A_b^\top.$$

This proposition is the domain-independent source of the memory-bounded streaming method developed below.

2.2 Central indefinite forms

The positive case is not the natural boundary of the central theory. Let

$$X \sim N(0, \Sigma), \qquad M = M^\top,$$

where M may be indefinite, and again write $S = \Sigma^{1/2}$. The symmetric operator

$$H = SMS$$

has a real signed spectrum.

Theorem 2.2 (Exact central signed reduction)**.** *If* $\lambda_1, \ldots, \lambda_r$ *are the nonzero eigenvalues of* H*, then*

$$\boxed{X^\top M X \overset{d}{=} \sum_{j=1}^{r} \lambda_j Z_j^2.}$$

Proof. Write $X = SZ$ with $Z \sim N(0, I)$ and diagonalize

$$SMS = V \Lambda V^\top.$$

Rotational invariance of the standard normal gives

$$Z^\top V \Lambda V^\top Z \overset{d}{=} \sum_j \lambda_j Z_j^2.$$

No positivity assumption on M is needed for this central reduction. □

The sign pattern determines the support. If every $\lambda_j \geq 0$, the lower endpoint is zero. If every $\lambda_j \leq 0$, the upper endpoint is zero. If both signs occur, the distribution is unbounded in both directions.

For the right-tail path, the cumulant generating function is

$$K(s) = -\frac{1}{2}\sum_j \log(1 - 2s\lambda_j),$$

with real domain

$$\boxed{\max_{\lambda_j<0} \frac{1}{2\lambda_j} < s < \min_{\lambda_j>0} \frac{1}{2\lambda_j}.}$$

When $q > E[Q] = \sum_j \lambda_j$, the saddlepoint is the unique positive solution of

$$K'(\theta) = q$$

inside this domain whenever a positive spectral component is present. The finite-upper-endpoint negative-only case is handled by the same tilted geometry with the endpoint fixed at zero.

The packet representation is sign aware: positive and negative eigenvalues are clustered separately, and each same-sign packet stores a center μ and centered moments. The local expansion remains

$$-\frac{1}{2}\log(1 - 2s(\mu + \delta)) = -\frac{1}{2}\log(1 - 2s\mu) + \frac{1}{2}\sum_{n\geq 1} \frac{\alpha(s)^n \delta^n}{n}, \qquad \alpha(s) = \frac{2s}{1 - 2s\mu},$$

with the same geometric remainder condition

$$|\alpha(s)| \max|\delta| < 1.$$

A useful distinction appears in the Fourier tail. Under the saddlepoint tilt,

$$|\phi_\theta(t)| = \prod_j \left[1 + \left(\frac{2\lambda_j t}{1 - 2\theta\lambda_j}\right)^2\right]^{-1/4}.$$

If r coefficients are nonzero, then asymptotically

$$|\phi_\theta(t)| = O(|t|^{-r/2}).$$

Thus certification difficulty depends strongly on spectral rank, not only on how small the requested probability is.

Exact finite-endpoint control. For

$$Q = -0.02\,\chi^2_{80},$$

the right tail is a lower chi-square tail with finite upper endpoint zero. The signed engine agrees with the exact distribution through

$$\boxed{p = 10^{-200}},$$

with maximum observed log-probability error about 2.84×10^{-14} decades and certificates around 10^{-12}.

Certified mixed-sign high-rank control. A deterministic rank-340 multiscale spectrum with 180 positive and 160 negative coefficients was tested from approximately 10^{-14} to

$$\boxed{p \approx 10^{-81.891937}}.$$

At the deepest point the packetized solver used 35 packets, $K = 4096$, and reported a relative certificate

$$5.78 \times 10^{-9}.$$

Its log-probability differed from the unpacketized full-spectrum signed solver by about 1.42×10^{-12} decades. Independent exponential-tilt importance sampling placed the discrepancy at about 1.41 Monte Carlo standard errors.

Retained low-rank limitation. The exact mixed-sign identity

$$Q = 0.07(\chi_2^2 - \chi_2^2)$$

is Laplace distributed with scale 0.14, so for $q \geq 0$,

$$P(Q \geq q) = \frac{1}{2} e^{-q/0.14}.$$

The computed values agree with this exact law through 10^{-200} to roughly 10^{-9} decades or better. However, the current analytic COS-series certificate rejects these low-rank mixed cases because the tilted characteristic function decays only as $|t|^{-2}$. The implementation reports this as a certificate failure rather than silently accepting the numerical value.

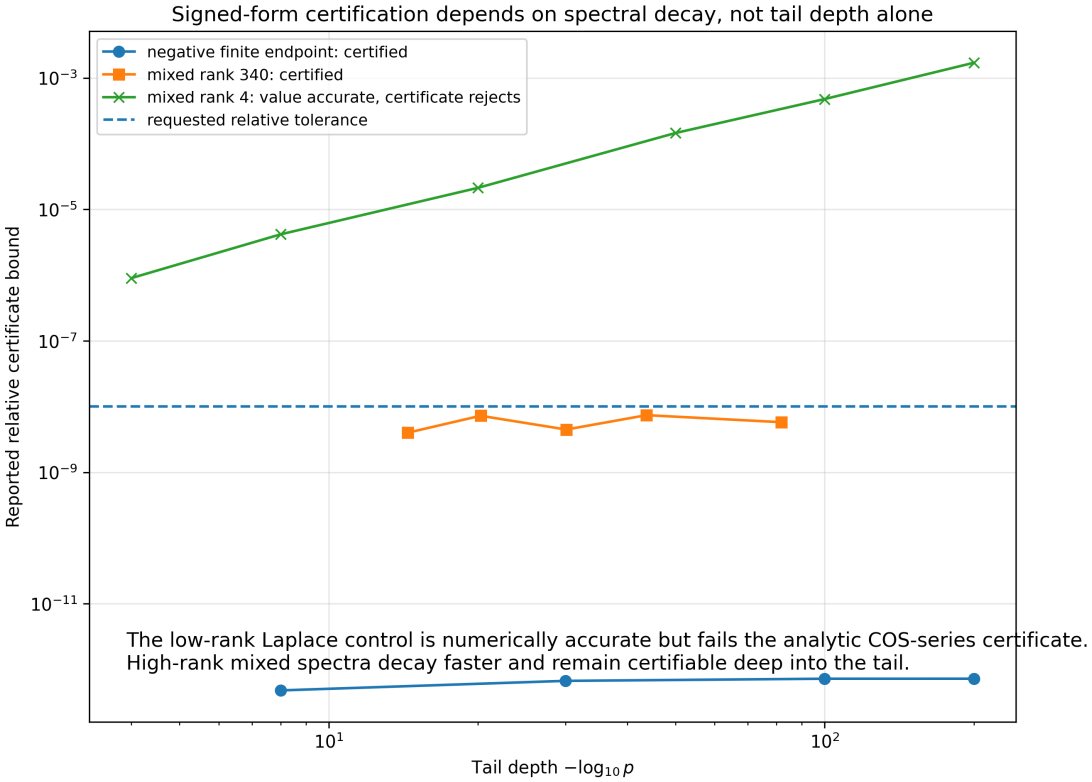


Figure 1: Signed-form certification boundary. Negative finite-endpoint forms and the rank-340 mixed spectrum remain certifiable in deep tails. The exact rank-4 Laplace control remains numerically accurate but fails the current analytic COS-series certificate because its Fourier decay is much slower.

2.3 Noncentral positive forms

Now let

$$X \sim N(\mu, \Sigma), \qquad M \succeq 0.$$

Again set $S = \Sigma^{1/2}$ and diagonalize

$$SMS = V \Lambda V^\top.$$

Writing

$$c = V^\top S M \mu,$$

each positive-eigenvalue direction satisfies

$$\lambda_i Y_i^2 + 2c_i Y_i = \lambda_i \left(Y_i + \frac{c_i}{\lambda_i} \right)^2 - \frac{c_i^2}{\lambda_i}.$$

Theorem 2.3 (Exact noncentral reduction)**.** *For the positive eigenvalues $\lambda_i > 0$,*

$$\boxed{Q \overset{d}{=} q_0 + \sum_i \lambda_i \chi_1^2(\delta_i),}$$

where

$$\delta_i = \left(\frac{c_i}{\lambda_i} \right)^2$$

and

$$q_0 = \mu^\top M \mu - \sum_i \frac{c_i^2}{\lambda_i} \geq 0.$$

If Σ is nonsingular on the M-active subspace, then $q_0 = 0$. When Σ is singular, q_0 is the deterministic quadratic contribution outside the random covariance support.

Proof. Expand

$$(\mu + SZ)^\top M (\mu + SZ) = Z^\top SMSZ + 2\mu^\top MSZ + \mu^\top M\mu$$

and rotate by V. Completing the square gives the displayed noncentral components. For a null vector of SMS,

$$M^{1/2} S v = 0,$$

hence

$$v^\top SM\mu = (M^{1/2} S v)^\top M^{1/2} \mu = 0,$$

so there is no unresolved stochastic linear term in a zero-eigenvalue direction. The remaining constant is the squared norm of the deterministic component and is therefore nonnegative. □

The production API implements this reduction directly. In an isotropic control,

$$\Sigma = I, \qquad M = aI,$$

the law collapses to the exact scaled noncentral chi-square distribution

$$Q/a \sim \chi_d^2(\|\mu\|^2).$$

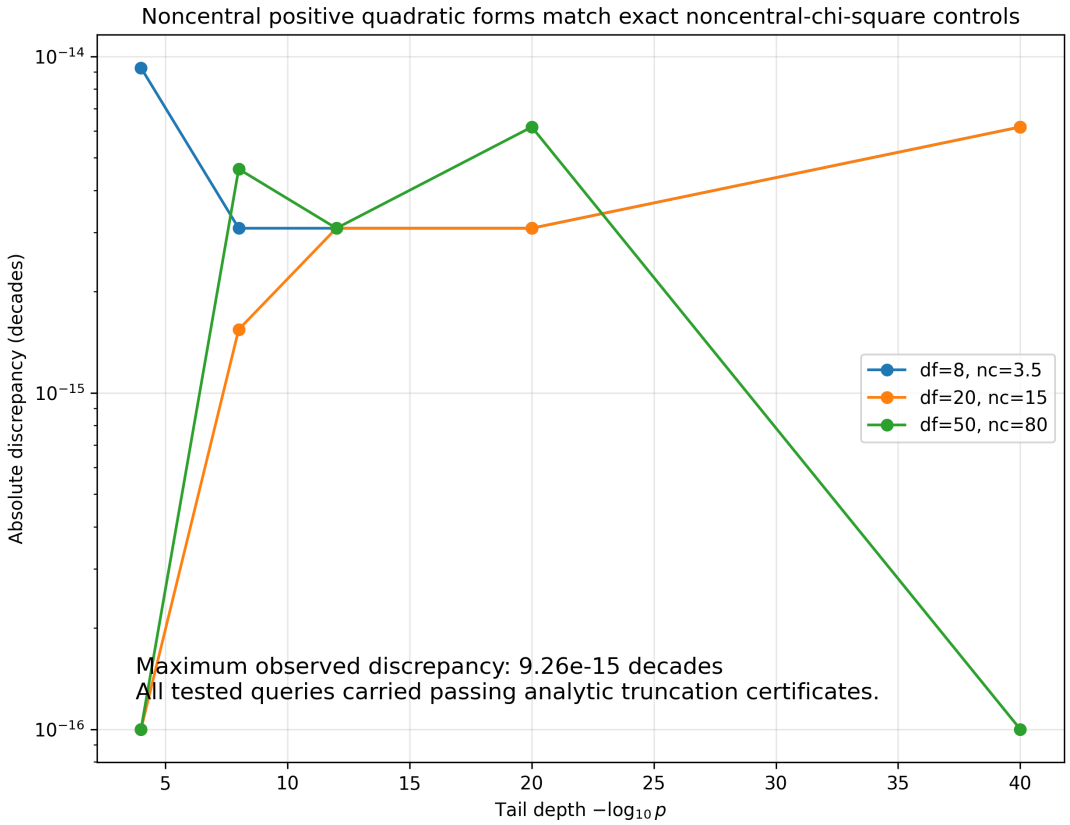


Figure 2: Independent noncentral controls. Three scaled isotropic Gaussian quadratic forms with different dimensions and noncentralities were compared against SciPy's noncentral-χ^2 law from 10^{-4} through 10^{-40}. The largest observed discrepancy was approximately 9.3×10^{-15} decades, with all analytic truncation certificates passing.

2.4 Decomposition axes

The solver separates four logically different representations:

$$\boxed{\text{storage blocks} \circ \text{structural phases} \circ \text{Krylov/range compression} \circ \text{spectral tail packets.}}$$

Storage blocks answer where data reside. Structural phases answer how the operator propagates. Krylov/range compression answers which subspace has actually been resolved. Spectral packets answer how the final transform is evaluated. These decompositions should not be conflated.

3 Demonstrated deep-tail precision

The method is designed for a regime in which direct survival probabilities can underflow long before the mathematical probability becomes numerically unresolvable. The engine therefore treats $\log p$ as the primary numerical quantity and keeps approximation certificates alive in the same extreme-tail regime.

The validation ladder spans seven qualitatively different settings:

validation regime	deepest $\log_{10} p$	observed error	certificate	spectral size
directed-rounding $0.02\chi^2_{80}$ control	-5000	enclosure width 3.37×10^{-11} decades	multiplicative ratio $\leq 1 + 7.75 \times 10^{-11}$	80 df
production-scale positive spectrum	-922.031162	below stored reference resolution	5.71×10^{-9}	rank 1997
corrected strong-dependence case study	-457.973916	below stored reference resolution	5.20×10^{-9}	rank 996
heterogeneous inversion benchmark near 10^{-16}	-16	2.88–6.52×10^{-15} decades	independent integral converged	ranks 80–120
signed finite-endpoint exact control	-200	$\leq 2.84 \times 10^{-14}$ decades	7.23×10^{-13}	rank 80
signed mixed multiscale	-81.891937	1.42×10^{-12} decades vs full grid	5.78×10^{-9}	rank 340
noncentral exact controls	-40	$\leq 9.26 \times 10^{-15}$ decades	passing	up to 50 df

The analytic scaled-chi-square ladder also includes

$$\log_{10} p = -100, -300, -1000, -5000,$$

with respective reported absolute log-probability discrepancies

$$2.84 \times 10^{-14}, \quad < 10^{-15}, \quad 3.41 \times 10^{-13}, \quad < 10^{-15}$$

in the stored benchmark output. The corresponding analytic relative certificates remain finite throughout, reaching only

$$3.85 \times 10^{-11}$$

at 10^{-5000}.

A production-scale run with $n = 2000$, $m = 50{,}000$, 49,951 retained features, and quadratic-form rank 1997 produced

$$\boxed{\log_{10} p = -922.0311624669905}$$

using 79 spectral packets and $K = 128$. Its independently evaluated full-spectrum tilted reference agreed within the stored reporting precision, while the reported relative certificate was

$$5.71 \times 10^{-9}.$$

The corrected strong-dependence case study independently reached

$$\boxed{\log_{10} p = -457.9739160722001}$$

with a finite 5.20×10^{-9} certificate. The signed finite-endpoint control remained certified through 10^{-200}, while a rank-340 mixed-sign multiscale spectrum remained certified to

$$\boxed{\log_{10} p = -81.89193686056221}.$$

Finally, the noncentral engine matched exact noncentral-chi-square controls from 10^{-4} through 10^{-40}, with maximum observed discrepancy

$$9.26 \times 10^{-15}$$

decades.

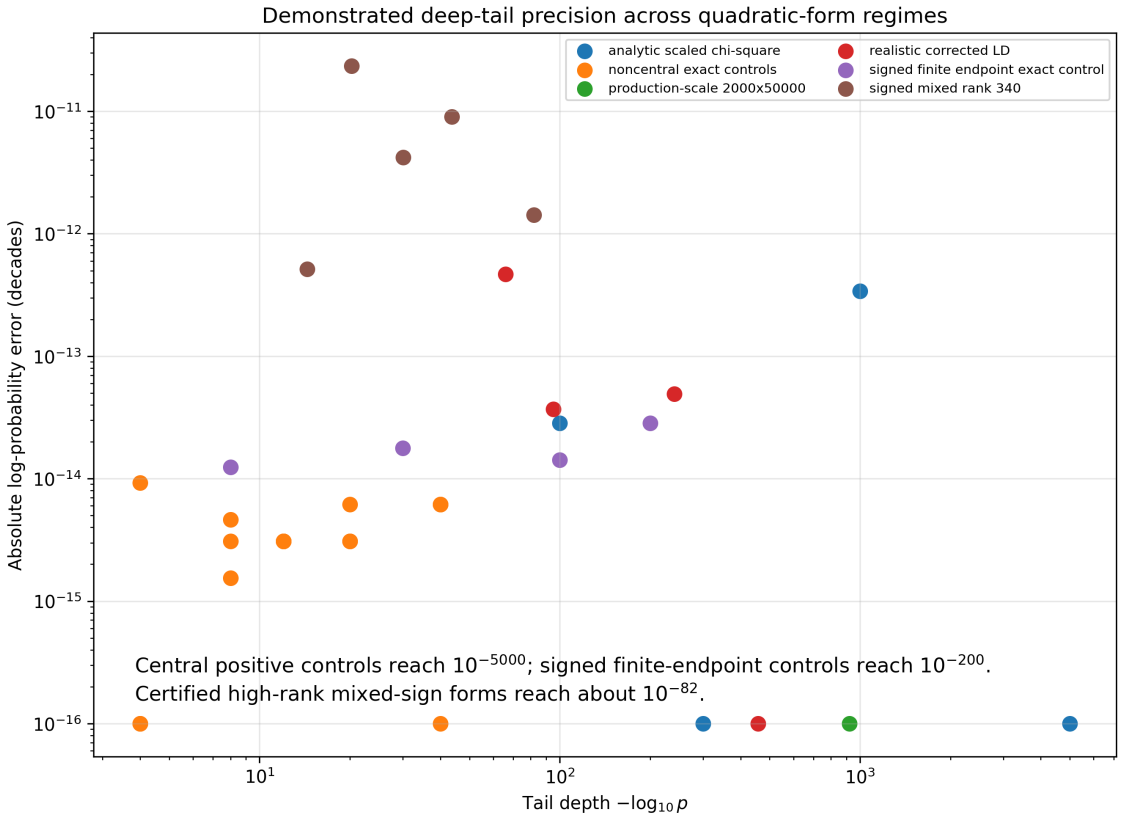


Figure 3: Demonstrated precision envelope across positive analytic controls, a production-scale rank-1997 spectrum, corrected strong-dependence spectra, signed finite-endpoint and mixed-sign forms, and noncentral exact controls. Points whose stored benchmark difference rounded to zero are placed at 10^{-16} decades only to make them visible on the logarithmic vertical axis.

3.1 Independent characteristic-function inversion near 10^{-16}

Closed-form controls are important, but they can hide numerical difficulties that appear only for heterogeneous spectra. An independent inversion test was therefore run on four generalized chi-square laws near

$$P(Q \geq q) \approx 10^{-16}.$$

The reference evaluates the original characteristic function directly through the central Imhof inversion identity

$$P(Q \geq q) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\sin \Theta(u)}{u\, R(u)}\, du,$$

where, for distinct positive coefficients λ_j with multiplicities h_j,

$$\Theta(u) = \frac{1}{2} \sum_j h_j \arctan(\lambda_j u) - \frac{1}{2} q u, \qquad R(u) = \prod_j \left(1 + \lambda_j^2 u^2\right)^{h_j/4}.$$

This route does not use exponential tilting, the COS expansion, spectral packets, or the K-COS error controller.

The four spectra include balanced, skewed, dominant-spike, and higher-rank cases. Each reference integral

was evaluated at two truncation limits. The omitted integral after U was bounded using

$$\frac{1}{\pi}\int_U^\infty \frac{du}{uR(u)} \le \frac{C}{\pi p}U^{-p}, \qquad p = \frac{1}{2}\sum_j h_j, \qquad C = \prod_j \lambda_j^{-h_j/2}.$$

spectrum	rank	$\lvert\Delta \log_{10} p\rvert$	relative difference	reference tail bound / p
balanced	80	6.52×10^{-15}	1.50×10^{-14}	6.84×10^{-20}
skewed	80	4.13×10^{-15}	9.50×10^{-15}	4.44×10^{-18}
spiky	80	5.19×10^{-15}	1.20×10^{-14}	7.00×10^{-17}
balanced	120	2.88×10^{-15}	6.64×10^{-15}	4.19×10^{-29}

The two inversion truncations agree to between approximately

$$1.2 \times 10^{-21} \quad \text{and} \quad 3.7 \times 10^{-17}$$

decades across the four cases. Thus the slow reference is numerically stable at a scale well below the observed K-COS-versus-inversion difference.

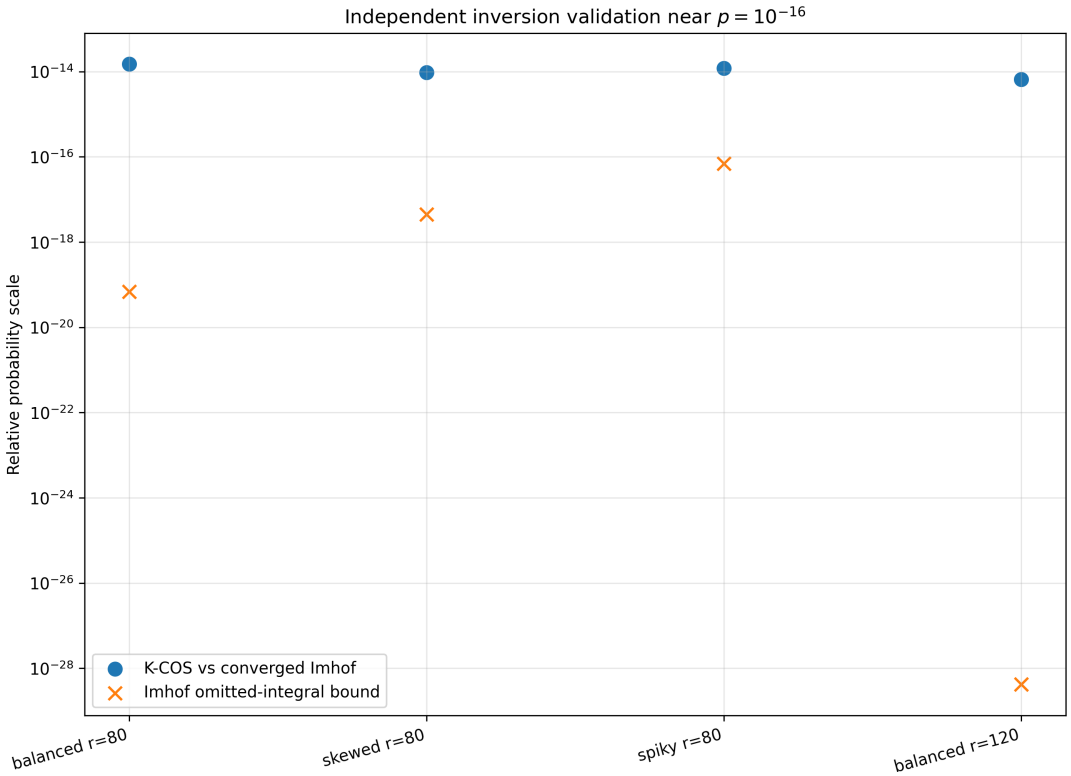


Figure 4: Independent direct-inversion validation near $p = 10^{-16}$ on four heterogeneous spectra. Circles show the relative K-COS/inversion difference; crosses show the analytic bound on the inversion integral omitted beyond the final truncation point.

This test is complementary to the exact scaled-chi-square controls. It asks whether the solver remains accurate when no single-gamma reduction is available and when an algorithmically different inversion of the full characteristic function is used as the reference.

These results have different evidentiary roles. The scaled-chi-square, finite-endpoint signed, Laplace signed, and noncentral cases provide analytic controls. The heterogeneous inversion battery, production-scale,

strong-dependence, and rank-340 mixed-sign cases test irregular spectra for which no single-gamma closed form is available. Together they test correctness against known distributions, representation compression, and stability on the intended high-rank numerical regime. The low-rank Laplace example also records a deliberate certificate failure boundary rather than conflating numerical agreement with certification.

4 Directed-rounding deep-tail enclosure

The preceding precision experiments combine analytic approximation bounds with stable floating-point evaluation. A stronger numerical statement is possible: return an interval that is guaranteed to contain the probability represented by an exact supplied positive spectrum.

This section gives the first such backend.

4.1 Interval target

For a positive central quadratic form

$$Q = \sum_{j=1}^{r} \lambda_j Z_j^2, \qquad \lambda_j > 0,$$

the output is not merely a point estimate. It is

$$\boxed{\log_{10} P(Q \geq q) \in [d_-, d_+].}$$

Hence the corresponding multiplicative uncertainty satisfies

$$\boxed{\frac{p_+}{p_-} \leq 10^{d_+ - d_-}.}$$

The finite transform and exponential prefactor are evaluated by directed-rounding interval arithmetic. The resulting interval is enlarged by two independently proved analytic remainders: a support/periodization alias bound and an omitted-COS-mode bound.

4.2 No exact saddlepoint is required

Choose any

$$0 < \theta < \frac{1}{2\lambda_{\max}}.$$

Exponential tilting gives

$$P(Q \geq q) = e^{K(\theta) - \theta q} S_\theta(q),$$

where

$$S_\theta(q) = E_\theta\left[e^{-\theta(Q-q)} 1_{\{Q \geq q\}}\right].$$

The interval proof does not require θ to be the exact saddlepoint. A high-precision saddlepoint is used only as an efficient design choice. Once serialized as an exact decimal input, it is simply a valid tilt inside the MGF domain.

The same is true of the support endpoint U and the auxiliary Chernoff parameter η. Correctness requires only

$$U > q, \qquad \theta < \eta < \frac{1}{2\lambda_{\max}}.$$

4.3 Rigorous support and alias control

Under the tilted measure,

$$P_\theta(Q > U) \le \exp\{K(\eta) - K(\theta) - (\eta - \theta)U\}.$$

The full-characteristic-function cosine expansion periodizes the tilted density on $[0, U]$. Because the original positive quadratic form is supported on $[0, \infty)$, the total positive-tail mass aliased back into one fundamental interval is at most $P_\theta(Q > U)$.

The actual payoff omitted above U is at most

$$e^{-\theta(U-q)} P_\theta(Q > U).$$

Therefore

$$\boxed{E_{\text{support}} \le P_\theta(Q > U)\left(1 + e^{-\theta(U-q)}\right).}$$

4.4 Rigorous infinite-mode control

Define

$$a_j = \frac{2\lambda_j}{1 - 2\theta\lambda_j}.$$

The normalized tilted characteristic function satisfies

$$|\phi_\theta(t)| = \prod_{j=1}^{r} \left(1 + (a_j t)^2\right)^{-1/4}.$$

Since

$$1 + (a_j t)^2 \ge (a_j t)^2,$$

we have

$$|\phi_\theta(t)| \le C_\theta t^{-r/2}, \qquad C_\theta = \prod_j a_j^{-1/2}.$$

For the payoff coefficient

$$I_k = \int_q^U e^{-\theta(x-q)} \cos(k\pi x/U)\, dx,$$

the elementary bound

$$|I_k| \le \frac{1}{\theta}$$

holds.

Let

$$p = \frac{r}{2} > 1.$$

For a finite transform retaining modes $0, \dots, K-1$,

$$\boxed{E_{\text{series}} \le \frac{2C_\theta}{U\theta}\left(\frac{U}{\pi}\right)^p \left[K^{-p} + \frac{K^{1-p}}{p-1}\right].}$$

This first interval backend therefore requires $r > 2$. Lower-rank positive forms can use a different tail bound and are not claimed by this theorem.

4.5 Final enclosure theorem

Let $\boldsymbol{S}_K$ be the directed-rounding interval containing the finite K-mode tilted COS sum, and set

$$E = E_{\text{support}} + E_{\text{series}}.$$

Then

$$\boxed{S_\theta(q) \in \boldsymbol{S}_K + [-E, E].}$$

If the lower endpoint of this interval is positive, directed rounding gives

$$\boxed{\log P(Q \geq q) \in K(\theta) - \theta q + \log\left(\boldsymbol{S}_K + [-E, E]\right).}$$

All finite arithmetic in this expression is enclosed by the interval backend.

4.6 Independent exact interval control through 10^{-5000}

For the control

$$Q = 0.02\chi^2_{80},$$

the survival probability has the finite exact formula

$$P(Q \geq q) = e^{-x} \sum_{j=0}^{39} \frac{x^j}{j!}, \qquad x = \frac{q}{0.04}.$$

This finite expression was evaluated independently by directed-rounding interval arithmetic, without using K-COS. The K-COS enclosure and this exact control enclosure overlap at

$$p = 10^{-100}, \qquad 10^{-1000}, \qquad 10^{-5000}.$$

With $K = 128$ and 90 interval decimal digits, the resulting K-COS widths are

target	width in $\log_{10} p$	multiplicative ratio minus one
10^{-100}	8.95×10^{-13}	2.06×10^{-12}
10^{-1000}	6.94×10^{-12}	1.60×10^{-11}
10^{-5000}	3.37×10^{-11}	7.75×10^{-11}

Thus the 10^{-5000} result is not merely a point estimate agreeing with a reference. For the exact supplied spectrum, it comes with a directed-rounding logarithmic enclosure whose probability ratio satisfies

$$\boxed{\frac{p_+}{p_-} \leq 1 + 7.75 \times 10^{-11}}$$

up to the displayed rounding of that ratio.

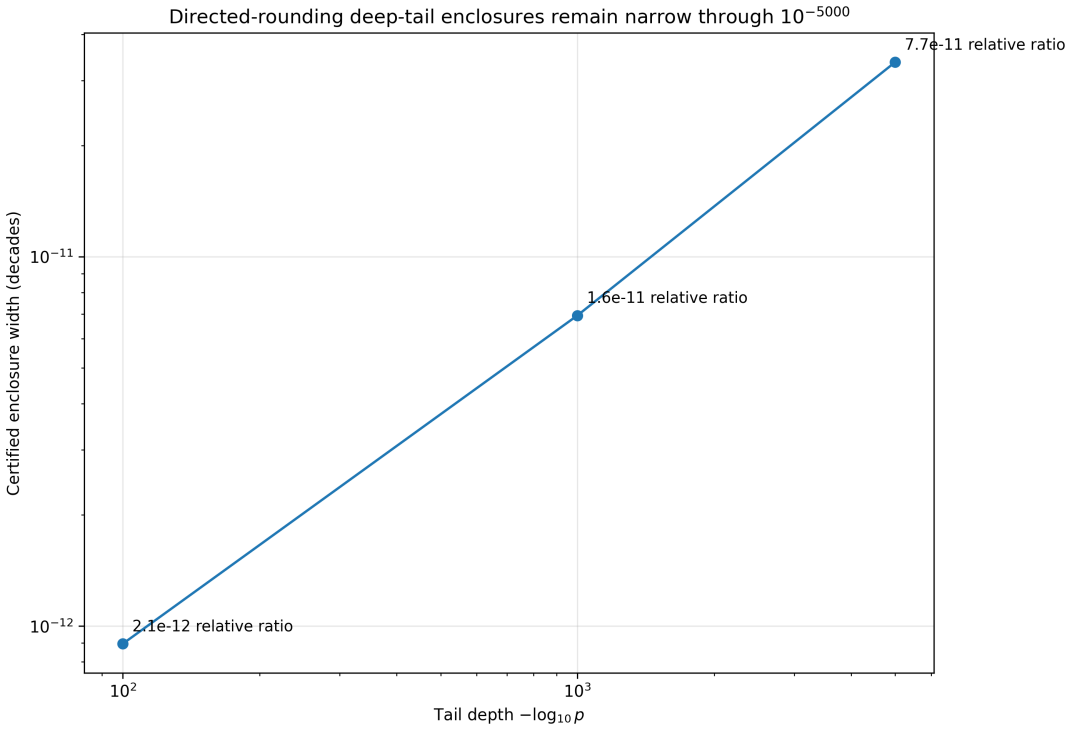


Figure 5: Width of the directed-rounding K-COS probability enclosure on exact scaled-chi-square controls. The enclosure remains narrow even at 10^{-5000}.

4.7 Arithmetic width versus analytic width

At the 10^{-5000} control, the interval precision was swept from 40 to 120 decimal digits. The total certified width remained essentially

$$3.34247 \times 10^{-11}$$

decades.

Over the same sweep, the arithmetic-only width contracted from approximately

$$7.5 \times 10^{-37}$$

to

$$6.3 \times 10^{-117}$$

decades.

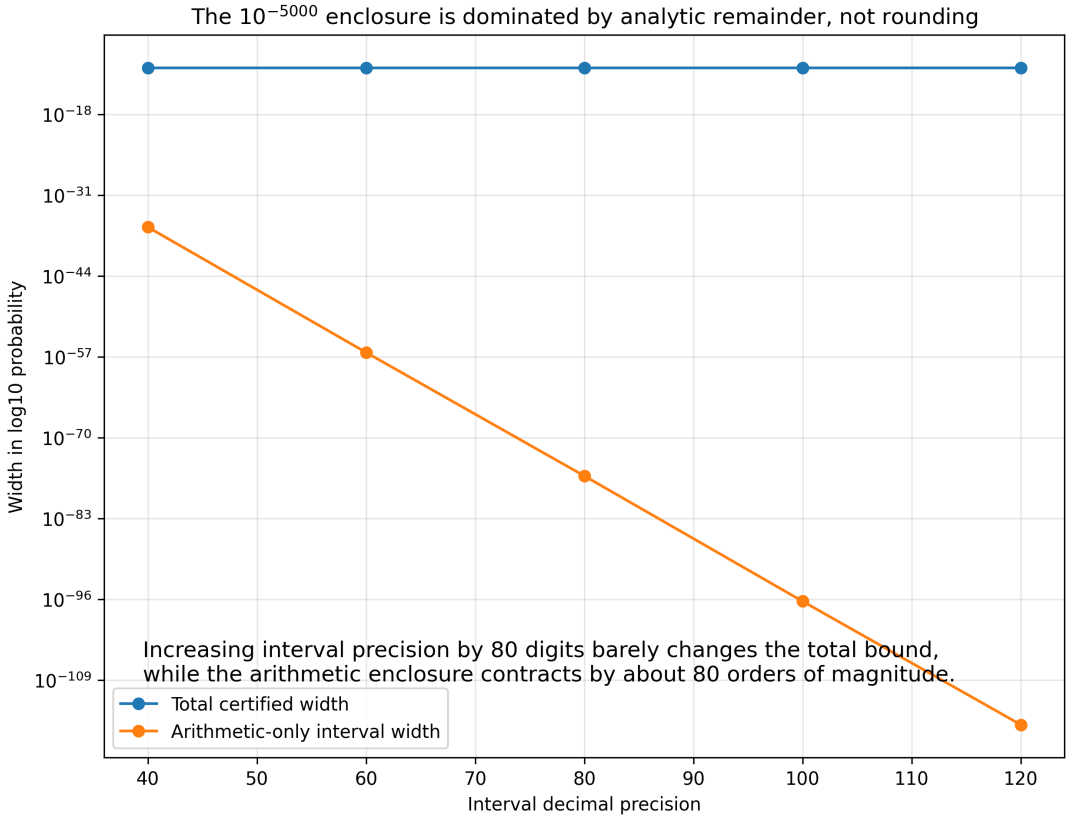


Figure 6: Arithmetic versus analytic uncertainty at a 10^{-5000} tail. The directed-rounding arithmetic enclosure becomes negligible; the final width is controlled by the explicit analytic support and series remainders.

This separation is important. Increasing numerical precision cannot improve an analytic truncation theorem. Conversely, once interval arithmetic is far below the analytic error budget, numerical rounding is no longer the limiting uncertainty.

4.8 Guarantee boundary

The v1.4 statement is intentionally narrow:

> For an exact supplied positive decimal spectrum of rank greater than two, the backend returns a directed-rounding logarithmic probability enclosure enlarged by proved analytic support/alias and omitted-mode bounds.

The result is conditional on correctness of the interval-arithmetic library implementation. The current implementation uses `mpmath.iv`, whose interval layer applies downward and upward endpoint rounding.

A completely end-to-end guarantee beginning from an arbitrary floating-point matrix requires one more layer: either a rigorously enclosed matrix-to-spectrum computation, or an exact/residual completion step whose linear algebra is also performed with verified interval or ball arithmetic.

The current packetized spectral representation has likewise not yet been ported to interval arithmetic. The first rigorous backend deliberately uses the exact supplied spectrum so that representation error cannot be hidden inside the numerical enclosure.

5 Accuracy-driven certification

The public numerical contract should specify the desired accuracy, not the internal transform resolution.

For an exact positive spectrum and threshold q, the solver accepts a requested relative multiplicative accuracy

$$\varepsilon_{\mathrm{req}} > 0$$

and returns a certified interval

$$[p_-, p_+]$$

only when

$$\boxed{\frac{p_+}{p_-} - 1 \le \varepsilon_{\mathrm{req}}.}$$

In logarithmic coordinates, if

$$\log_{10} p \in [d_-, d_+],$$

the acceptance rule is

$$\boxed{10^{d_+ - d_-} - 1 \le \varepsilon_{\mathrm{req}}.}$$

No value of K, support width, or arithmetic precision appears in the external contract.

Theorem 5.1 (Accuracy contract for the positive exact-spectrum backend)**.** *Assume an exact supplied positive decimal spectrum of rank greater than two, a threshold q, and requested multiplicative tolerance $\varepsilon_{\mathrm{req}} > 0$. Suppose the directed-rounding interval backend returns an enclosure*

$$\log_{10} p \in [d_-, d_+]$$

that includes the proved support/alias and omitted-mode remainders. If the accuracy-driven controller returns `certified=True`*, then*

$$\boxed{P(Q \ge q) \in [10^{d_-}, 10^{d_+}]}$$

and

$$\boxed{\frac{10^{d_+}}{10^{d_-}} - 1 \le \varepsilon_{\mathrm{req}}.}$$

If the resource caps are reached before this inequality is proved, the controller returns `certified=False` *rather than promoting the interval midpoint to a certified answer.*

Proof. The directed-rounding backend encloses the finite transform and exponential prefactor and then enlarges that enclosure by the proved support/alias and infinite-mode error bounds. Therefore the true tail probability lies inside the returned interval. The controller evaluates $\mathrm{expm1}(\log(10)(d_+ - d_-))$ at explicit high precision and sets `certified=True` only when the resulting upper ratio is no larger than $\varepsilon_{\mathrm{req}}$. The fail-closed branch follows directly from the controller definition. □

5.1 Adaptive refinement

The current controller adjusts three internal resources:

1. the COS mode count K;
2. the tilted support scale L, which changes U and its associated Chernoff parameter;

3. the interval arithmetic precision.

At each stage the full v1.4 directed-rounding interval is recomputed. If the rounding component is material relative to the requested budget, arithmetic precision is increased. Otherwise the dominant analytic contribution determines the next refinement: mode error increases K, while support/alias error increases L.

The solver terminates only after the multiplicative inequality above has been proved.

5.2 Fail-closed behavior

If the configured resource caps are exhausted before the requested accuracy is established, the solver does not silently return its midpoint as though it were certified.

Instead it returns

`certified = false`

together with the unresolved interval and a failure reason.

This behavior is part of the numerical definition of the production solver.

5.3 Accuracy ladder through 10^{-5000}

The exact control

$$Q = 0.02\chi^2_{80}$$

was evaluated at tail depths

$$10^{-100}, \qquad 10^{-1000}, \qquad 10^{-5000}.$$

For each depth the controller was asked to prove relative multiplicative accuracies

$$10^{-8}, \quad 10^{-10}, \quad 10^{-12}, \quad 10^{-14}.$$

All twelve contracts were satisfied.

At the 10^{-5000} point, the controller produced:

requested accuracy	achieved ratio minus one	K	L
10^{-8}	3.19×10^{-10}	64	10
10^{-10}	7.70×10^{-11}	128	10
10^{-12}	3.48×10^{-15}	128	12
10^{-14}	3.48×10^{-15}	128	12

The gap between requested and achieved accuracy is expected: refinement is discrete, so one additional support or mode refinement can improve the certificate by several orders of magnitude.

A further stress test requested

$$10^{-16} \quad \text{and} \quad 10^{-18}$$

relative accuracy at the same 10^{-5000} tail. The solver obtained approximately

$$3.05 \times 10^{-18}$$

and

$$9.20 \times 10^{-20}$$

respectively.

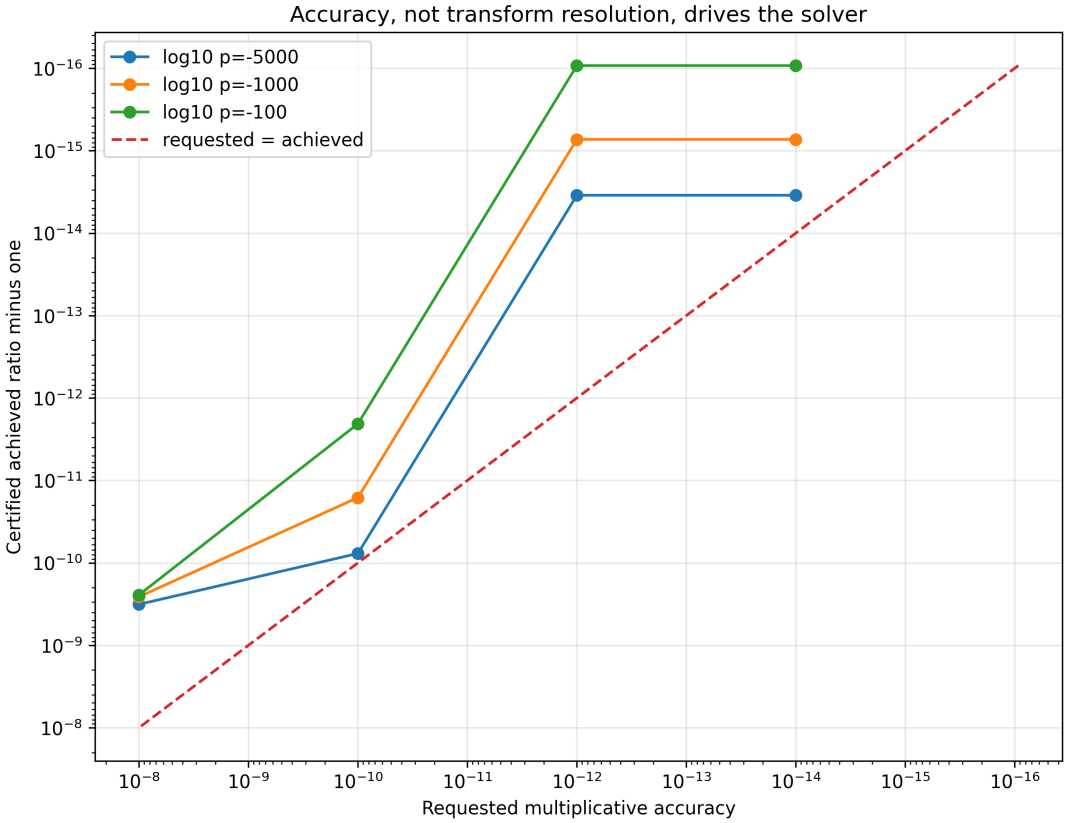


Figure 7: Accuracy-driven refinement. Each point lies at or below the requested-equals-achieved line because a result is labeled certified only after the requested multiplicative probability accuracy has been proved.

5.4 Acceptance arithmetic is itself high precision

The acceptance condition must be evaluated as carefully as the probability interval.

An early implementation converted a tiny log-width w using

$$10^{w} - 1$$

under the default multiprecision context. At approximately 10^{-15} relative scales this could round the ratio to zero.

The enclosure itself remained valid, but that conversion was not acceptable for a certification decision. The production implementation now evaluates

$$\boxed{\operatorname{expm1}(\log(10)\, w)}$$

inside an explicit high-precision context before deciding whether the requested accuracy has been met.

This regression is retained because the correctness requirement applies not only to the tail integral but also to the logic that labels the answer certified.

5.5 Input semantics at $p \approx 10^{-16}$

At this accuracy scale, the representation of the threshold q itself is measurable. In the $0.02\chi^2_{80}$ control targeted at 10^{-16}, rounding the high-precision threshold to binary float64 shifts the exact tail by approximately

$$3.83 \times 10^{-15}$$

decades. The hardened float64 solver evaluated at that same rounded input then differs from the exact tail of the rounded input by approximately

$$5.17 \times 10^{-15}$$

decades.

Accordingly, claims below the float64 input-quantization scale use exact-decimal or interval inputs. The rigorous accuracy contract therefore certifies the mathematical inputs supplied to it, rather than silently identifying a rounded binary threshold with an ideal decimal threshold.

6 Verified matrix-to-spectrum bridge

The strongest tail result above begins from a positive spectrum. This section moves the rigor boundary one level upstream by certifying the spectrum produced from a floating-point factor matrix.

6.1 Binary64 input semantics

Let

$$A \in \mathbb{R}^{n\times m}$$

be supplied as an IEEE binary64 array. Each stored entry is treated as its exact dyadic rational value. The target operator is therefore the exact Gram matrix

$$H = AA^T$$

defined by those binary64 values, not an unspecified nearby real matrix.

A high-precision symmetric eigensolver is used only to generate candidate quantities

$$Q \in \mathbb{R}^{n\times n}, \qquad D = \operatorname{diag}(d_1, \ldots, d_n), \qquad d_1 \geq \cdots \geq d_n \geq 0.$$

No correctness claim depends on trusting this eigensolver.

6.2 Post-certified spectral enclosure theorem

Theorem 6.1 (Candidate eigensystem post-certification)**.** *Let*

$$B = QDQ^T.$$

Suppose directed-rounding computation proves

$$\|Q^TQ - I\|_2 \leq \eta < 1$$

and

$$\|H - B\|_2 \leq \rho.$$

Then the ordered eigenvalues of the positive semidefinite matrix H satisfy

$$\boxed{\max\{0,(1-\eta)d_i-\rho\}\le\lambda_i(H)\le(1+\eta)d_i+\rho}$$

for every i.

Proof. Set

$$S=Q^TQ.$$

The first hypothesis gives the Loewner inequalities

$$(1-\eta)I\preceq S\preceq(1+\eta)I.$$

The eigenvalues of

$$B=(QD^{1/2})(QD^{1/2})^T$$

are the same as those of

$$D^{1/2}SD^{1/2}.$$

Hence

$$(1-\eta)D\preceq D^{1/2}SD^{1/2}\preceq(1+\eta)D,$$

and eigenvalue monotonicity yields

$$(1-\eta)d_i\le\lambda_i(B)\le(1+\eta)d_i.$$

Weyl's perturbation inequality and $\|H-B\|_2\le\rho$ then give

$$|\lambda_i(H)-\lambda_i(B)|\le\rho.$$

Combining the two bounds and using $H\succeq 0$ proves the claim. □

The implementation does not attempt to evaluate the spectral norms above exactly. Instead it encloses every entry of

$$Q^TQ-I\quad\text{and}\quad H-QDQ^T$$

with directed-rounding interval arithmetic and uses

$$\|M\|_2\le\|M\|_F.$$

Thus the reported η and ρ are conservative certified upper bounds.

6.3 Monotone propagation into the tail

Let the spectral theorem produce

$$\ell_i\le\lambda_i(H)\le u_i.$$

Using the same independent standard normals in all three sums gives

$$\sum_i\ell_iZ_i^2\le\sum_i\lambda_iZ_i^2\le\sum_i u_iZ_i^2\qquad\text{almost surely.}$$

Therefore

$$\boxed{P\left(\sum_i\ell_iZ_i^2\ge q\right)\le P(Q\ge q)\le P\left(\sum_i u_iZ_i^2\ge q\right).}$$

The lower endpoint is evaluated by the rigorous positive-tail backend on the lower spectrum, and the upper endpoint by the same backend on the upper spectrum. The final matrix-defined probability interval is accepted only when its multiplicative ratio satisfies the requested accuracy.

6.4 Random dense spectral-containment audit

Three unrelated dense binary64 factors were tested against independently recomputed high-precision eigenvalues of their exact dyadic Gram matrices:

factor size	all eigenvalues contained	maximum relative width	time (s)
12×18	yes	3.36×10^{-58}	0.09
20×30	yes	1.44×10^{-57}	0.35
28×36	yes	1.20×10^{-56}	0.80

No reference eigenvalue fell outside its certified interval.

6.5 End-to-end matrix-defined tail near 10^{-16}

A dense 64×64 exact-binary factor was constructed from a normalized Hadamard basis with four singular-value groups. The stored binary64 entries are exact dyadic rationals. The spectral postcheck returned

$$\eta \le 1.65 \times 10^{-49}, \qquad \rho \le 9.18 \times 10^{-50}.$$

For example, the leading coefficient was enclosed by

$$0.999\ldots97435 < \lambda_1 < 1.000\ldots02565,$$

with a width of order 10^{-49}.

At

$$q = 131.33201485100517,$$

the tail lies near 10^{-16}. Starting from the factor matrix itself:

requested relative accuracy	achieved ratio minus one	certified
10^{-10}	2.60×10^{-12}	yes
10^{-12}	4.09×10^{-15}	yes

For the 10^{-12} request, the final enclosure was approximately

$$\log_{10} p \in [-15.99999999999996625, -15.99999999999996447].$$

This closes the basic numerical chain

$$\boxed{\text{binary64 factor} \longrightarrow \text{certified spectrum} \longrightarrow \text{certified deep-tail interval}}$$

at moderate matrix dimension.

6.6 Scalability boundary

The theorem is general, but the present verifier forms a full candidate basis Q and postchecks dense matrices. It therefore closes the correctness boundary before it closes the large-scale memory boundary. For very large streamed factors, the next step is to replace the full-basis postcheck with a verified range-core/residual certificate and interval exact-completion step, so that the matrix-to-spectrum guarantee inherits the same memory-bounded architecture developed elsewhere in this manuscript.

7 Weighted-feature specialization

A prominent application constructs the general factor from weighted, covariate-residualized features. A rare-variant kernel statistic is one concrete instance, but the same formulas apply to any feature matrix with nonnegative column weights.

Let

$$G \in \mathbb{R}^{n\times m}, \qquad W = \mathrm{diag}(w_1, \ldots, w_m), \quad w_j \geq 0,$$

and let $Q_X \in \mathbb{R}^{n\times p}$ have orthonormal columns spanning the null-model covariates. Define

$$M_X = I - Q_X Q_X^\top, \qquad A = M_X G W^{1/2}, \qquad K = AA^\top.$$

The central implementation difficulty is not the algebraic size of the model but the mismatch between the total factor size $n \times m$ and the working memory of an accelerator. The key point of this manuscript is that several decompositions act on different mathematical coordinates and therefore compose:

storage blocks ∘ structural phases ∘ Krylov/range compression ∘ spectral tail packets.

Storage blocking controls residence. Structural phases encode LD windows, annotations, frequency bands, or other coordinate groups. Krylov methods compress the operator to the rank actually requested. Spectral packets compress the final generalized-χ^2 transform.

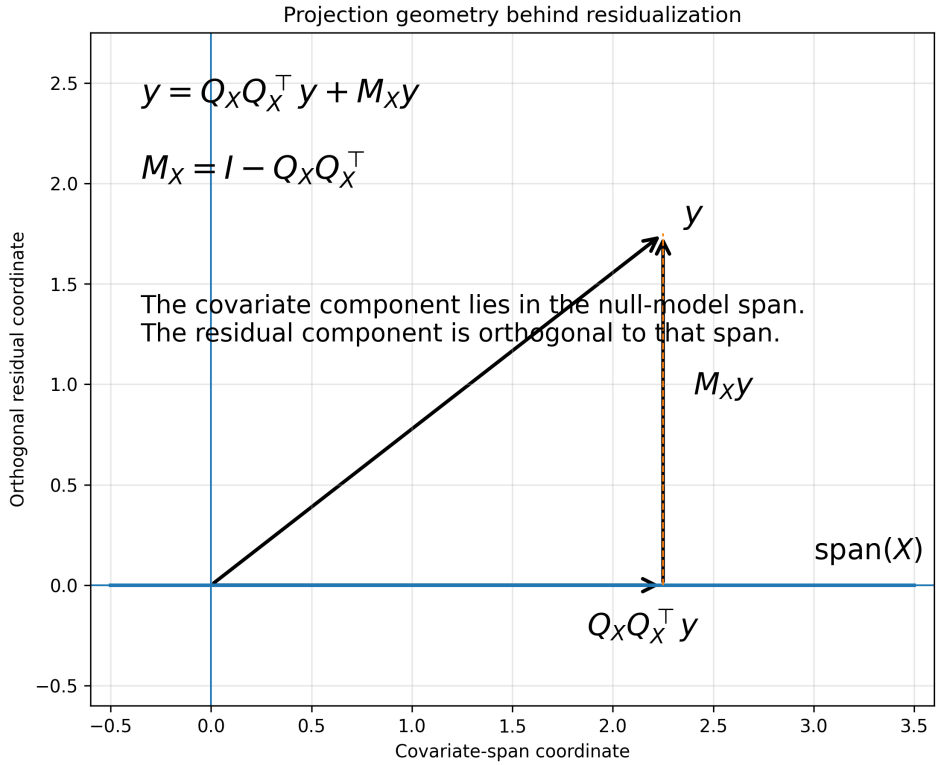


Figure 8: Residualization is the geometric starting point. Every sample-space vector splits into a component in the covariate span and an orthogonal residual component. The factor $A = M_X G W^{1/2}$ therefore acts only on variation remaining after the null-model covariates have been removed.

8 Variant-block Gram decomposition

Partition the variant indices into disjoint blocks

$$\{1, \dots, m\} = J_1 \sqcup \cdots \sqcup J_B.$$

Let

$$G_b = G_{[:,J_b]}, \qquad W_b = W_{J_b,J_b}, \qquad A_b = M_X G_b W_b^{1/2}.$$

Then

$$A = [A_1 \ \cdots \ A_B].$$

Theorem 8.1 (Additive block Gram identity)**.**

$$\boxed{K = AA^\top = \sum_{b=1}^{B} A_b A_b^\top = M_X \left(\sum_{b=1}^{B} G_b W_b G_b^\top \right) M_X.}$$

Proof. Horizontal concatenation gives

$$[A_1 \ \cdots \ A_B] \begin{bmatrix} A_1^\top \\ \vdots \\ A_B^\top \end{bmatrix} = \sum_b A_b A_b^\top.$$

The final form follows from $A_b = M_X G_b W_b^{1/2}$ and $M_X^\top = M_X = M_X^2$. □

Remark 8.2. The theorem is exact for every partition. Block boundaries are execution parameters, not statistical parameters.

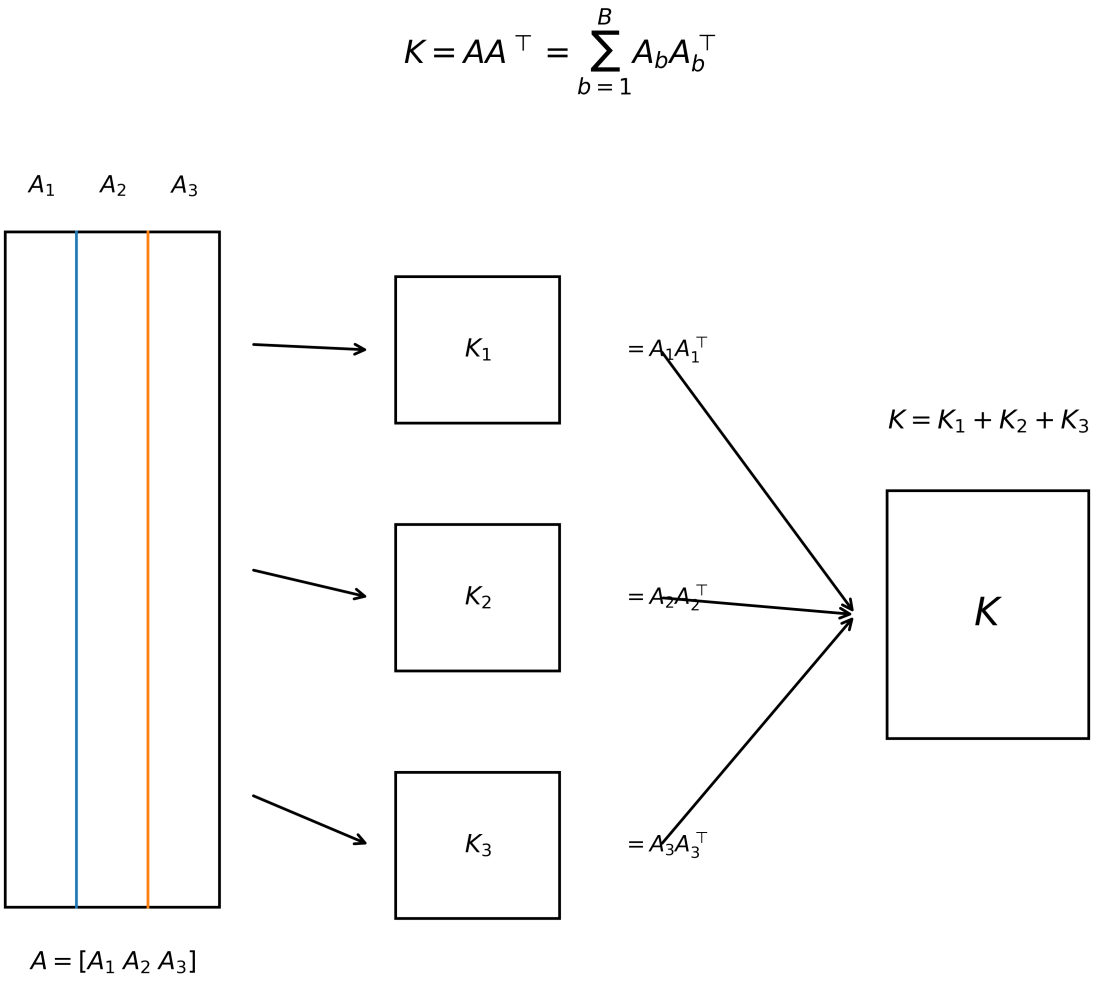


Figure 9: Exact block-kernel assembly. The sample interaction kernel carried by each resident variant block is $K_b = A_b A_b^\top$, and the full kernel is their exact sum. Streaming changes the order of evaluation, not the operator.

9 Exact streamed kernel actions

Theorem 9.1 (One-pass sample-space streaming action)**.** *For every block vector $V \in \mathbb{R}^{n\times s}$,*

$$\boxed{KV = M_X \sum_{b=1}^{B} G_b W_b G_b^\top M_X V.}$$

Proof. Apply the additive block Gram identity to KV, then associate each block product as

$$G_b W_b (G_b^\top M_X V).$$

□

Algorithm 9.2 (Streamed KV)**.** 1. Compute $U = M_X V$.

2. Set $Y = 0$.

3. For each resident block b, compute
$$T_b = G_b^\top U, \qquad Y \leftarrow Y + G_b(W_b T_b).$$
4. Return $M_X Y$.

Corollary 9.3 (Working memory independent of total variant count)**.** *If at most c variants are resident and the block-vector width is s, then the algebraic working set is*
$$\boxed{O(nc + ns + cs + np),}$$
independent of total variant count m.

10 Streamed factor actions and Golub–Kahan

Let $V \in \mathbb{R}^{m\times s}$ be partitioned conformally:
$$V = \begin{bmatrix} V_1 \\ \vdots \\ V_B \end{bmatrix}.$$

Theorem 10.1 (Streamed AV)**.**
$$\boxed{AV = M_X \sum_{b=1}^{B} G_b W_b^{1/2} V_b.}$$

Theorem 10.2 (Streamed $A^\top U$)**.** *For $U \in \mathbb{R}^{n\times s}$,*
$$\boxed{(A^\top U)_b = W_b^{1/2} G_b^\top M_X U.}$$

Proof. Both identities follow by substituting $A = M_X G W^{1/2}$ and partitioning the variant coordinate. □

Corollary 10.3 (Streaming Golub–Kahan equivalence)**.** *Any exact-arithmetic algorithm whose access to A occurs only through AV and $A^\top U$ produces the same iterates when those products are implemented by the preceding streamed formulas.*

This applies directly to block Golub–Kahan bidiagonalization, matrix-free singular-value iteration, range finding, and restarted variants. The factor need not be resident.

11 Two-pass dual action

The variant-space normal operator is
$$H = A^\top A.$$

Theorem 11.1 (Two-pass streamed dual action)**.** *For every $V \in \mathbb{R}^{m\times s}$,*
$$\boxed{HV = A^\top (AV).}$$
The product is evaluated exactly by a streamed pass forming $Y = AV \in \mathbb{R}^{n\times s}$, followed by a second streamed pass emitting the blocks
$$(HV)_b = W_b^{1/2} G_b^\top M_X Y.$$

This form is particularly natural for variant-space polyphase Krylov iterations because the right-coordinate blocks can be produced and consumed sequentially.

12 Two-dimensional sample/variant tiling

A variant block may itself exceed device memory. Let

$$H_b = G_b W_b^{1/2}$$

and partition samples into row tiles $I_1, \dots, I_R$.

Theorem 12.1 (Two-pass residualized tile construction)**.** *Define*

$$R_b = Q_X^\top H_b.$$

Then

$$\boxed{R_b = \sum_r Q_{X,I_r}^\top H_{b,I_r},}$$

and after R_b is known,

$$\boxed{A_{b,I_r} = H_{b,I_r} - Q_{X,I_r} R_b.}$$

Consequently,

$$\boxed{K_{I_r,I_s} = \sum_b A_{b,I_r} A_{b,I_s}^\top.}$$

Proof. The first identity is the row-partitioned matrix product $Q_X^\top H_b$. The second is the definition $M_X H_b = H_b - Q_X Q_X^\top H_b$. The last identity is the corresponding sample-tile expansion of $AA^\top$. □

With sample tile height t and resident variant width c, the dominant working storage is

$$O(tc + pc + t^2).$$

13 Partition refinement and permutation invariance

Theorem 13.1 (Partition-refinement invariance)**.** *Let $\mathcal{J}$ be any partition of the variants and let $\mathcal{J}'$ refine $\mathcal{J}$. Then*

$$\boxed{\sum_{J \in \mathcal{J}} A_J A_J^\top = \sum_{J' \in \mathcal{J}'} A_{J'} A_{J'}^\top = AA^\top.}$$

The analogous streamed operator action is also invariant.

Proof. Each coarse block is a disjoint union of refined blocks. Apply the additive Gram theorem within each coarse block and sum. □

Corollary 13.2 (Adaptive memory splitting)**.** *A runtime may recursively subdivide any block that exceeds available device memory without changing the exact operator.*

Proposition 13.3 (Variant permutation invariance)**.** *If the columns of G and the diagonal entries of W are permuted by the same permutation, K and every streamed sample-space action are unchanged.*

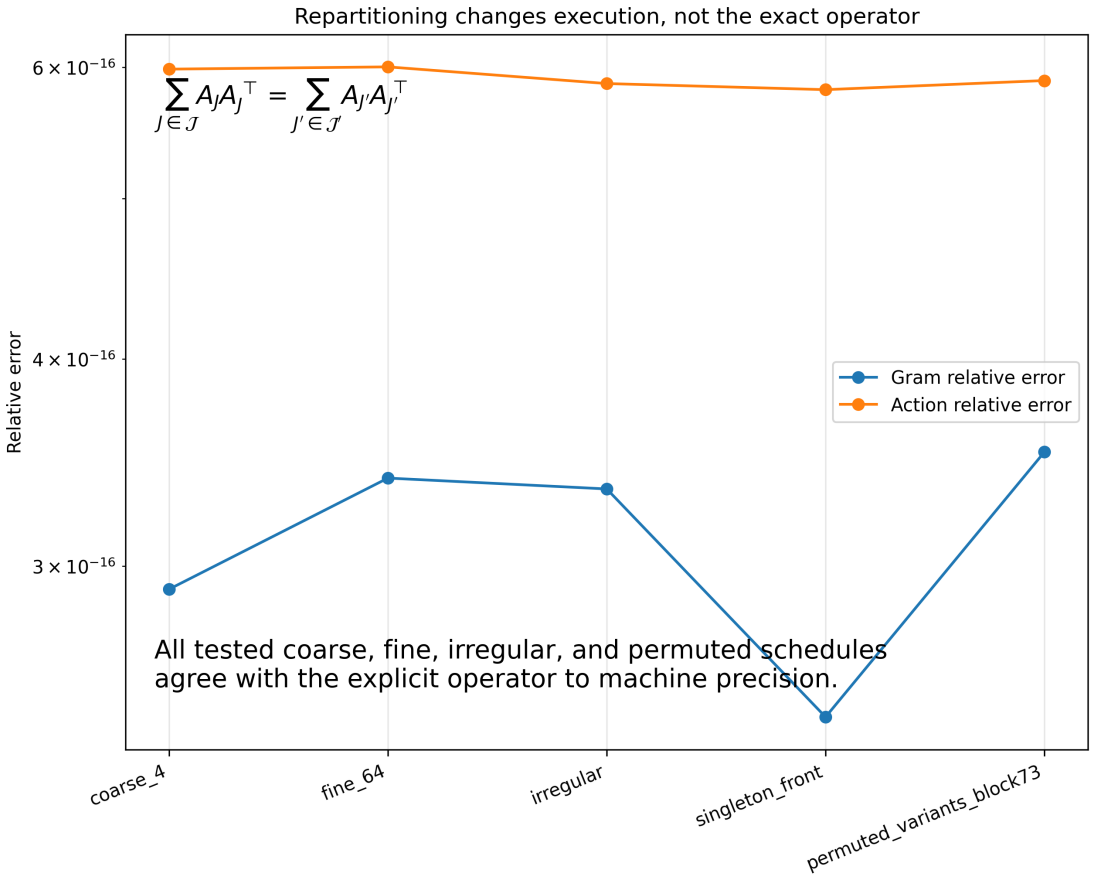


Figure 10: Partition-refinement invariance in floating-point tests. Coarse, fine, irregular, singleton-heavy, and permuted blockings reproduce the explicitly formed Gram matrix and its action to machine precision. The block schedule is therefore an execution choice rather than a change in the statistical model.

14 Polyphase structure

Let P_h project onto variant phase h, and again set

$$H = A^\top A.$$

Define the exact phase graph by

$$g \to h \quad \Longleftrightarrow \quad P_h H P_g \neq 0.$$

Theorem 14.1 (Polyphase light cone)**.** *If the graph distance from phase g to phase h exceeds k, then*

$$\boxed{P_h H^k P_g = 0.}$$

Proof. Insert the projector resolution $\sum_j P_j = I$ between successive factors of H. Every nonzero term in $P_h H^k P_g$ corresponds to a phase path of length k. If no such path exists, every term vanishes. □

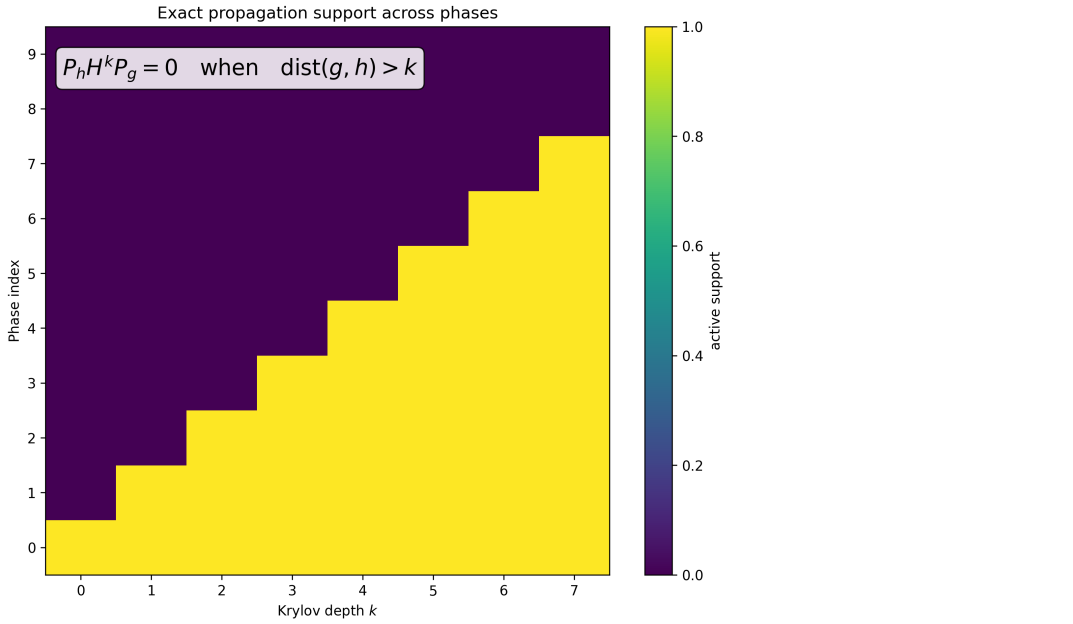


Figure 11: The exact polyphase light cone. The support of the Krylov iterate expands by at most one graph edge per operator application. Phases outside graph distance k remain identically inactive at depth k and need not be loaded.

Theorem 14.2 (Streaming-polyphase composition)**.** *At Krylov depth k, an exact matrix-free iteration may omit every phase outside the exact light cone. Within each active phase, arbitrary storage refinement may be used.*

Proof. The light-cone theorem removes inactive phase contributions exactly. Partition-refinement invariance shows that the remaining active phase operators may be streamed through any exact subpartition. □

This is the key composition law:

$$\boxed{\text{mathematical locality} \longrightarrow \text{active phases} \longrightarrow \text{active storage blocks.}}$$

15 Certified approximate light cones

The exact light-cone theorem requires literal zero phase couplings. Real LD, annotation, and feature graphs are often only approximately sparse. This section replaces exact support by a computable phasewise envelope.

Let the phase projectors be $P_1, \ldots, P_q$, and let X be a vector or block vector. Define its phase-norm profile

$$\rho(X)_g = \|P_g X\|_F,$$

and define the nonnegative phase-transfer matrix

$$\mathcal{B}_{hg} = \|P_h H P_g\|_2.$$

The matrix $\mathcal{B}$ records the strength of every one-step phase transfer without assuming that weak couplings vanish.

Theorem 15.1 (Weighted phase-transfer envelope)**.** *For every block vector X,*

$$\boxed{\rho(HX) \le \mathcal{B}\rho(X)}$$

componentwise. Consequently, for every integer $k \ge 0$,

$$\boxed{\rho(H^k X) \le \mathcal{B}^k \rho(X).}$$

Proof. Using $\sum_g P_g = I$ and submultiplicativity,

$$\begin{aligned}\|P_h H X\|_F &= \left\| \sum_g P_h H P_g X \right\|_F \\ &\le \sum_g \|P_h H P_g\|_2 \|P_g X\|_F \\ &= (\mathcal{B}\rho(X))_h.\end{aligned}$$

Iteration gives the power bound. □

Now choose a symmetric set of retained phase blocks and define

$$H_0 = \sum_{(h,g)\in\mathcal{E}_0} P_h H P_g.$$

Let

$$(\mathcal{B}_0)_{hg} = \begin{cases} \mathcal{B}_{hg}, & (h,g) \in \mathcal{E}_0, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathcal{R} = \mathcal{B} - \mathcal{B}_0.$$

Symmetric pair retention preserves self-adjointness of H_0, although the following error theorem does not require positive semidefiniteness of H_0.

Theorem 15.2 (Thresholded-power certificate)**.** *For every $k \ge 1$,*

$$\boxed{\rho((H^k - H_0^k)X) \le (\mathcal{B}^k - \mathcal{B}_0^k)\rho(X).}$$

In particular,

$$\boxed{\|(H^k - H_0^k)X\|_F \le \left\|(\mathcal{B}^k - \mathcal{B}_0^k)\rho(X)\right\|_2.}$$

Proof. Write $E = H - H_0$. The telescoping identity gives

$$H^k - H_0^k = \sum_{j=0}^{k-1} H^{k-1-j} E H_0^j.$$

The block-norm matrix of E is $\mathcal{R}$. Applying Theorem 15.1 successively to each term gives

$$\rho((H^k - H_0^k)X) \le \sum_{j=0}^{k-1} \mathcal{B}^{k-1-j} \mathcal{R} \mathcal{B}_0^j \rho(X).$$

The matrix telescoping identity

$$\mathcal{B}^k - \mathcal{B}_0^k = \sum_{j=0}^{k-1} \mathcal{B}^{k-1-j} (\mathcal{B} - \mathcal{B}_0) \mathcal{B}_0^j$$

yields the phasewise formula. Orthogonality of the phase projectors gives

$$\|Y\|_F^2 = \sum_h \|P_h Y\|_F^2,$$

which proves the global bound. □

Corollary 15.3 (Certified approximate light cone)**.** *Assume X is supported on a set of start phases S. If phase h is not reachable from S by a retained path of length at most k, then*

$$P_h H_0^k X = 0$$

and

$$\boxed{\|P_h H^k X\|_F \le \left[(\mathcal{B}^k - \mathcal{B}_0^k)\rho(X)\right]_h.}$$

Thus an exact zero outside the retained light cone is replaced by a certified nonzero leakage budget.

Corollary 15.4 (Scalar threshold law)**.** *For any induced matrix norm, let*

$$M = \max\{\|\mathcal{B}\|, \|\mathcal{B}_0\|\}.$$

Then

$$\boxed{\|\mathcal{B}^k - \mathcal{B}_0^k\| \le k M^{k-1} \|\mathcal{R}\|.}$$

Hence weak discarded coupling produces a first-order error bound proportional to the discarded transfer norm and to the Krylov depth.

Theorem 15.5 (Krylov-polynomial certificate)**.** *Let*

$$p(t) = \sum_{j=0}^{d} a_j t^j.$$

Then

$$\boxed{\rho\big((p(H) - p(H_0))X\big) \le \sum_{j=1}^{d} |a_j| (\mathcal{B}^j - \mathcal{B}_0^j)\rho(X).}$$

Proof. Apply Theorem 15.2 to every nonconstant power and use the triangle inequality. The constant terms cancel. □

The polynomial theorem covers arbitrary vectors in a finite Krylov space, including polynomial filters and restarted combinations, rather than only the monomial $H^k X$.

Corollary 15.6 (Certified phase pruning)**.** *If a thresholded phase graph is chosen so that*

$$\left\| (\mathcal{B}^k - \mathcal{B}_0^k)\rho(X) \right\|_2 \le \tau,$$

then replacing $H^k X$ by $H_0^k X$ incurs Frobenius error at most τ. The same statement holds for a Krylov polynomial with the certificate in Theorem 15.5.

This turns graph thresholding into an error-budget decision. A phase or interface is not discarded because it merely appears small; it is discarded only when the weighted sum of all paths using discarded couplings remains inside the requested tolerance.

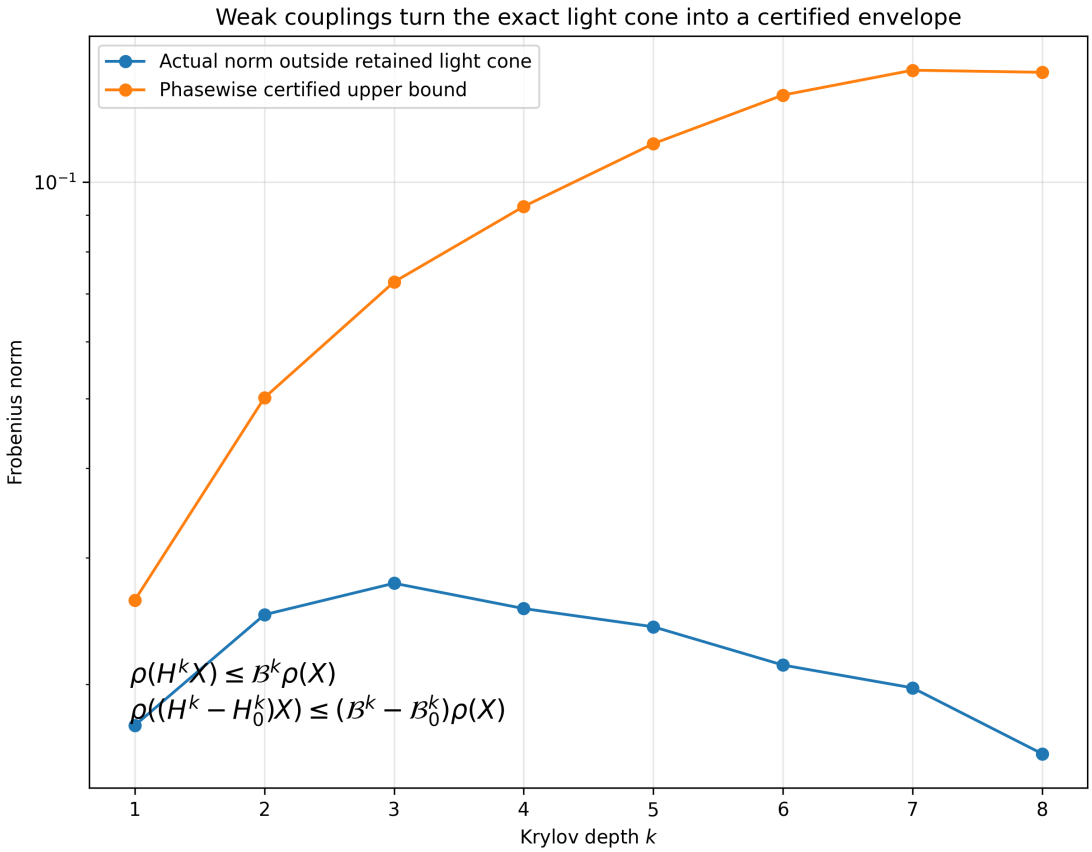


Figure 12: Weak long-range phase couplings replace the exact zero outside the retained light cone by a nonzero but certified envelope. In this PSD test, the phasewise transfer-matrix certificate bounds the observed leakage at every depth from one through eight.

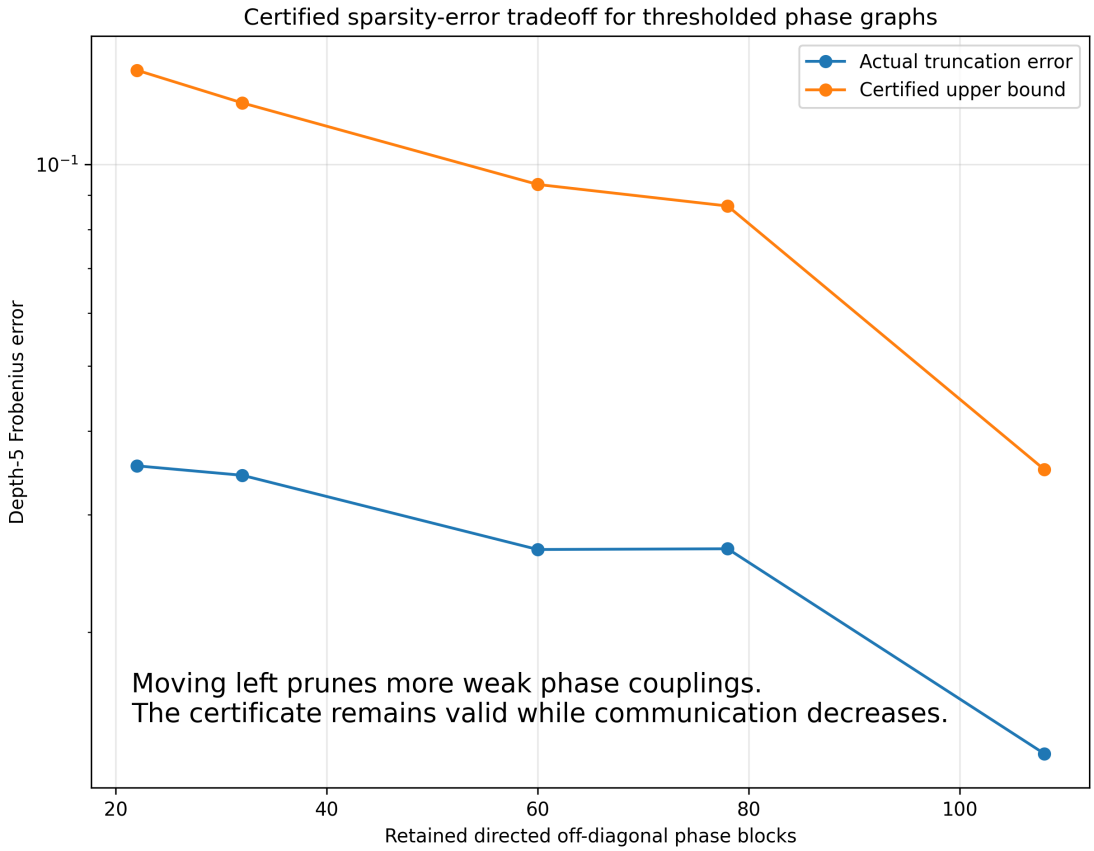


Figure 13: Certified phase-graph sparsification at Krylov depth five. Retaining fewer off-diagonal phase blocks reduces communication and operator work while the transfer-matrix bound remains above the measured truncation error. The bound is intentionally conservative because it ignores cancellation and singular-vector alignment.

15.1 Numerical audit

The validation operator has twelve phases of width four. A block-bidiagonal factor generates a positive-definite local operator H_0, and weak symmetric long-range blocks are then added to obtain H. Both normalized operators remain positive definite. A three-column start block is supported in phase zero.

For depths one through eight:

- every observed phase norm lay below $\mathcal{B}^k\rho(X)$;
- every observed thresholding error lay below $(\mathcal{B}^k - \mathcal{B}_0^k)\rho(X)$;
- the largest total observed-error-to-certificate ratio was approximately 0.669.

A separate hard-threshold sweep retained between 22 and 132 directed off-diagonal phase blocks. Every measured depth-five error remained below its certificate; the largest observed-error-to-certificate ratio was approximately 0.376. Finally, a degree-four signed polynomial in H satisfied the phasewise polynomial certificate in all twelve phases, with maximum ratio approximately 0.068.

These computations validate the implementation and illustrate sharpness. The universal result is supplied by the preceding proofs, not by the finite test.

16 Certified budget-optimal phase scheduling

The preceding section answers a validation question: given a retained phase operator H_0, how large can the Krylov error be? This section turns that certificate into a selection rule. The goal is to remove as much phase communication as possible while remaining inside a prescribed error budget.

Let $\mathcal{E}$ denote the set of undirected off-diagonal phase pairs. Because H is self-adjoint, $\mathcal{B}$ is symmetric. For a phase pair $e = \{h, g\}$ define the nonnegative matrix Δ_e by

$$(\Delta_e)_{hg} = \mathcal{B}_{hg}, \qquad (\Delta_e)_{gh} = \mathcal{B}_{gh},$$

with every other entry zero. If $\mathcal{O} \subseteq \mathcal{E}$ is the set of omitted pairs, define

$$\mathcal{B}_{\mathcal{O}} = \mathcal{B} - \sum_{e \in \mathcal{O}} \Delta_e.$$

The diagonal is always retained.

For depth $k \geq 1$ and starting phase profile $\rho_0 = \rho(X)$, define the *edge risk vector*

$$\boxed{q_e^{(k)} = \sum_{j=0}^{k-1} \mathcal{B}^{k-1-j} \Delta_e \mathcal{B}^j \rho_0}$$

and its scalar risk

$$\boxed{\beta_e^{(k)} = \|q_e^{(k)}\|_2.}$$

Unlike the raw block norm $\mathcal{B}_{hg}$, this quantity depends on the start profile and on Krylov depth.

Theorem 16.1 (Edge-additive leakage majorant)**.** *For every omitted set $\mathcal{O} \subseteq \mathcal{E}$,*

$$\boxed{0 \leq (\mathcal{B}^k - \mathcal{B}_{\mathcal{O}}^k)\rho_0 \leq \sum_{e \in \mathcal{O}} q_e^{(k)}}$$

componentwise. Consequently,

$$\boxed{\|(H^k - H_{\mathcal{O}}^k)X\|_F \leq \left\|(\mathcal{B}^k - \mathcal{B}_{\mathcal{O}}^k)\rho_0\right\|_2 \leq \sum_{e \in \mathcal{O}} \beta_e^{(k)}.}$$

Proof. The first inequality in the second display is Theorem 15.2. For the transfer matrices, the exact noncommutative telescoping identity is

$$\mathcal{B}^k - \mathcal{B}_{\mathcal{O}}^k = \sum_{j=0}^{k-1} \mathcal{B}^{k-1-j}(\mathcal{B} - \mathcal{B}_{\mathcal{O}})\mathcal{B}_{\mathcal{O}}^j.$$

All matrices and ρ_0 are nonnegative, and $0 \leq \mathcal{B}_{\mathcal{O}} \leq \mathcal{B}$ entrywise. Hence

$$\mathcal{B}_{\mathcal{O}}^j \rho_0 \leq \mathcal{B}^j \rho_0.$$

Substituting

$$\mathcal{B} - \mathcal{B}_{\mathcal{O}} = \sum_{e \in \mathcal{O}} \Delta_e$$

and distributing the sum gives the componentwise majorant. The final scalar bound follows from the Euclidean triangle inequality. □

Corollary 16.2 (Polynomial edge-risk certificate)**.** *Let*

$$p(t) = \sum_{d=0}^{D} a_d t^d.$$

Define

$$q_e^{(p)} = \sum_{d=1}^{D} |a_d| q_e^{(d)}, \qquad \beta_e^{(p)} = \|q_e^{(p)}\|_2.$$

Then

$$\boxed{\|(p(H) - p(H_{\mathcal{O}}))X\|_F \le \sum_{e \in \mathcal{O}} \beta_e^{(p)}.}$$

Thus the same scheduling machinery applies to finite Krylov combinations and polynomial filters.

Corollary 16.3 (Query-invisible phase pair)**.** *If*

$$\Delta_e \mathcal{B}^j \rho_0 = 0 \qquad \text{for every } 0 \le j < k,$$

then

$$\beta_e^{(k)} = 0.$$

Therefore a phase pair can be large in operator norm and still have zero risk for a particular finite-depth query. A magnitude-only threshold cannot represent this distinction.

16.1 Equal-cost optimality

Suppose each retained phase pair has the same communication or scheduling cost. Under the additive certificate, a set of omitted pairs is feasible if

$$\sum_{e \in \mathcal{O}} \beta_e^{(k)} \le \tau.$$

Sort the risks as

$$\beta_{(1)} \le \beta_{(2)} \le \cdots \le \beta_{(N)}.$$

Theorem 16.4 (Cardinality-optimal certified pruning)**.** *Let r be the largest integer satisfying*

$$\sum_{i=1}^{r} \beta_{(i)} \le \tau.$$

Then omitting the r least-risk phase pairs is feasible, and no feasible omitted set can contain more than r pairs under the scalar additive certificate.

Proof. Among all subsets of cardinality $r+1$, the smallest possible sum of risks is

$$\sum_{i=1}^{r+1} \beta_{(i)}.$$

By maximality of r, that sum exceeds τ. Hence no set of $r+1$ omitted pairs is feasible. The prefix of length r is feasible by construction. □

This is a genuine optimization theorem, not a heuristic threshold rule. The optimization target is the additive certificate itself; the tighter exact transfer envelope can leave additional slack.

16.2 Exact-envelope refinement

After the cardinality-optimal additive schedule is found, recomputing

$$\left\|(\mathcal{B}^k - \mathcal{B}^k_{\mathcal{O}})\rho_0\right\|_2$$

may reveal unused budget. Additional pairs can then be removed one at a time, provided the exact transfer envelope remains below τ.

Proposition 16.5 (Certified refinement)**.** *Any sequence of additional omissions accepted only when*

$$\left\|(\mathcal{B}^k - \mathcal{B}^k_{\mathcal{O}})\rho_0\right\|_2 \le \tau$$

remains a valid τ-certificate for the original operator error.

The refinement rule used in the reference implementation chooses, among all currently feasible next omissions, the pair that produces the smallest increase in the exact transfer envelope. It is certified but is not claimed globally optimal.

16.3 Unequal communication costs

Let $c_e \ge 0$ denote the communication or storage cost saved by omitting pair e. Under the scalar additive certificate, the exact discrete selection problem is

$$\boxed{\max_{x_e\in\{0,1\}} \sum_e c_e x_e \quad \text{subject to} \quad \sum_e \beta_e^{(k)} x_e \le \tau.}$$

This is the 0–1 knapsack problem. Hence unequal-cost scheduling is combinatorial in general. The implementation supplies a safe cost-to-risk greedy heuristic; its output remains certified because it never exceeds the additive risk budget. A fractional-knapsack relaxation supplies an upper bound on the maximum possible saved cost.

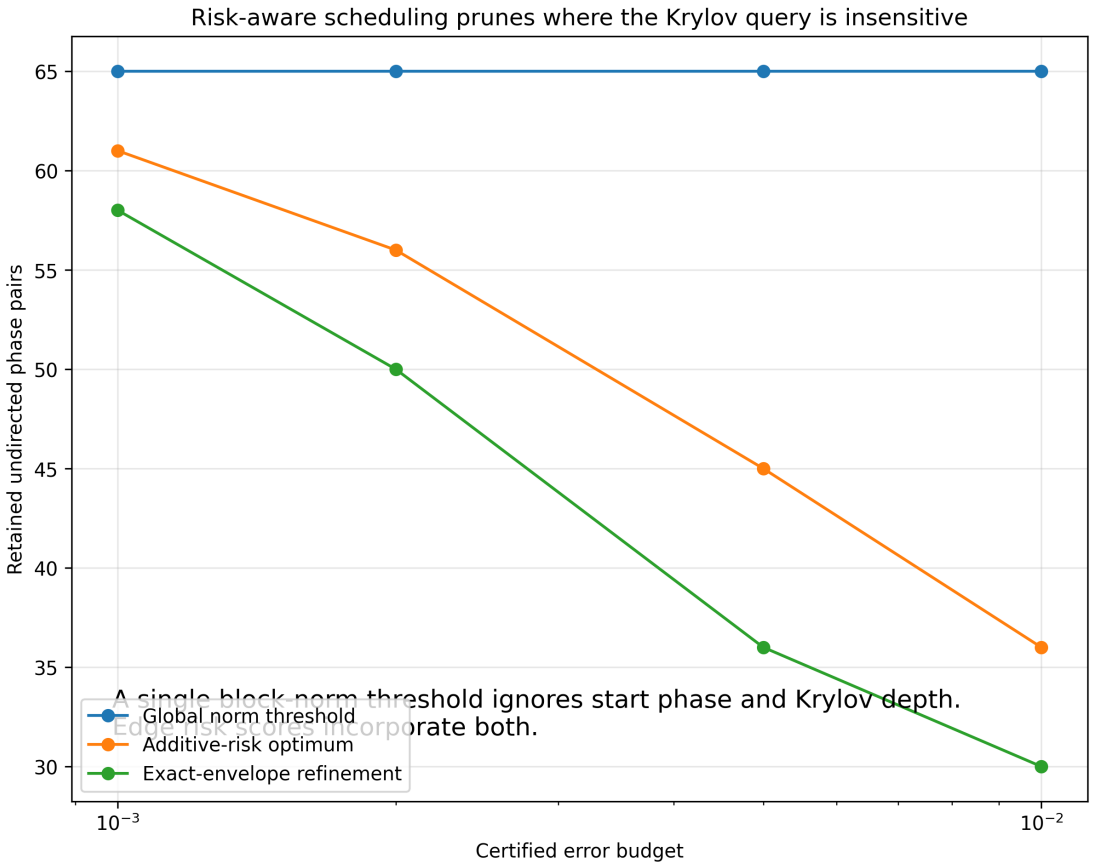


Figure 14: Depth-five pruning under a fixed certified error budget. A global block-norm threshold is nearly unable to prune this example without crossing the requested tolerance, whereas query-aware edge risks remove many phase pairs. The exact-envelope refinement safely uses slack left by the additive majorant.

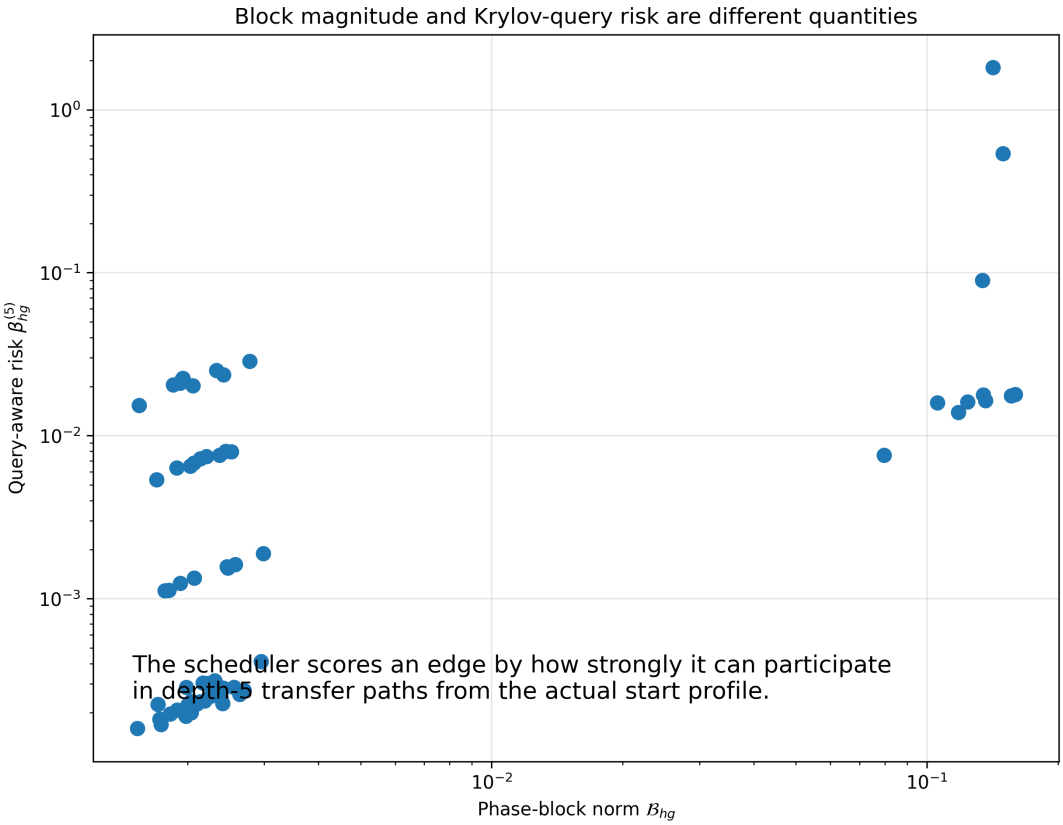


Figure 15: Raw phase-block magnitude and finite-depth query risk are different quantities. The edge score $\beta_{hg}^{(5)}$ incorporates the start profile and every depth-five transfer path through the phase graph; the block norm $\mathcal{B}_{hg}$ does not.

16.4 Numerical audit

The twelve-phase operator from Section 15 was reused at Krylov depth five. There are 66 undirected off-diagonal phase pairs. For error budgets 10^{-3}, 2×10^{-3}, 5×10^{-3}, and 10^{-2}, the additive-optimal scheduler retained respectively 61, 56, 45, and 36 pairs. Exact-envelope refinement reduced those counts further to 58, 50, 36, and 30 pairs while remaining inside budget.

In the same test, the best single global threshold on the raw block norm retained 65 pairs at every one of those four budgets. The difference is structural: the global threshold is blind to start phase and Krylov depth.

A separate six-phase problem contains 15 candidate phase pairs, so all $2^{15} = 32768$ subsets can be enumerated. At four distinct budgets, the sorted-risk scheduler matched the brute-force cardinality optimum under the additive certificate in every case. The exact-envelope greedy refinement also matched the brute-force exact-envelope optimum in cardinality for all four budgets in this finite audit. The latter agreement is empirical and is not promoted to a theorem.

For an unequal-cost eight-phase test, costs were proportional to the bytes in symmetric dense phase blocks. The cost-to-risk heuristic saved 5408 bytes out of a 9472-byte communication proxy while using additive risk 0.011632 under budget 0.012. The fractional relaxation upper bound was 5647.8 bytes, placing the greedy schedule within about 4.25% of that relaxation bound on this instance.

17 Asynchronous double-buffer execution model

The mathematical streaming identities do not require transfers and GEMMs to be serial. Consider B storage blocks. Let t_b be the transfer time for block b and c_b its device compute time.

Proposition 17.1 (Ideal two-buffer pipeline)**.** *In a model with perfect asynchronous overlap, transfer of block $b+1$ may run while block b is being processed. The serial one-buffer time is*

$$T_{\text{serial}} = \sum_{b=1}^{B} (t_b + c_b),$$

whereas the two-buffer pipeline time is

$$\boxed{T_{\text{pipe}} = t_1 + \sum_{b=1}^{B-1} \max\{c_b, t_{b+1}\} + c_B.}$$

For uniform times $t_b = t$ and $c_b = c$,

$$T_{\text{pipe}} = t + c + (B-1)\max\{t, c\}.$$

Proof. After the first block arrives, every interior pipeline stage finishes only when both the current computation and the next transfer have completed. Their joint stage duration is therefore the maximum of those two times. The initial transfer and final compute cannot be hidden and form the two boundary terms. □

If one resident block uses M_{block} bytes and all other fixed working state uses M_{fixed} bytes, the corresponding memory models are

$$M_{\text{single}} = M_{\text{fixed}} + M_{\text{block}}, \qquad \boxed{M_{\text{double}} = M_{\text{fixed}} + 2M_{\text{block}}.}$$

The proposition is a scheduling model, not a hardware timing claim. PCIe or NVLink contention, kernel launch overhead, synchronization, page pinning, and library workspaces can reduce realized overlap.

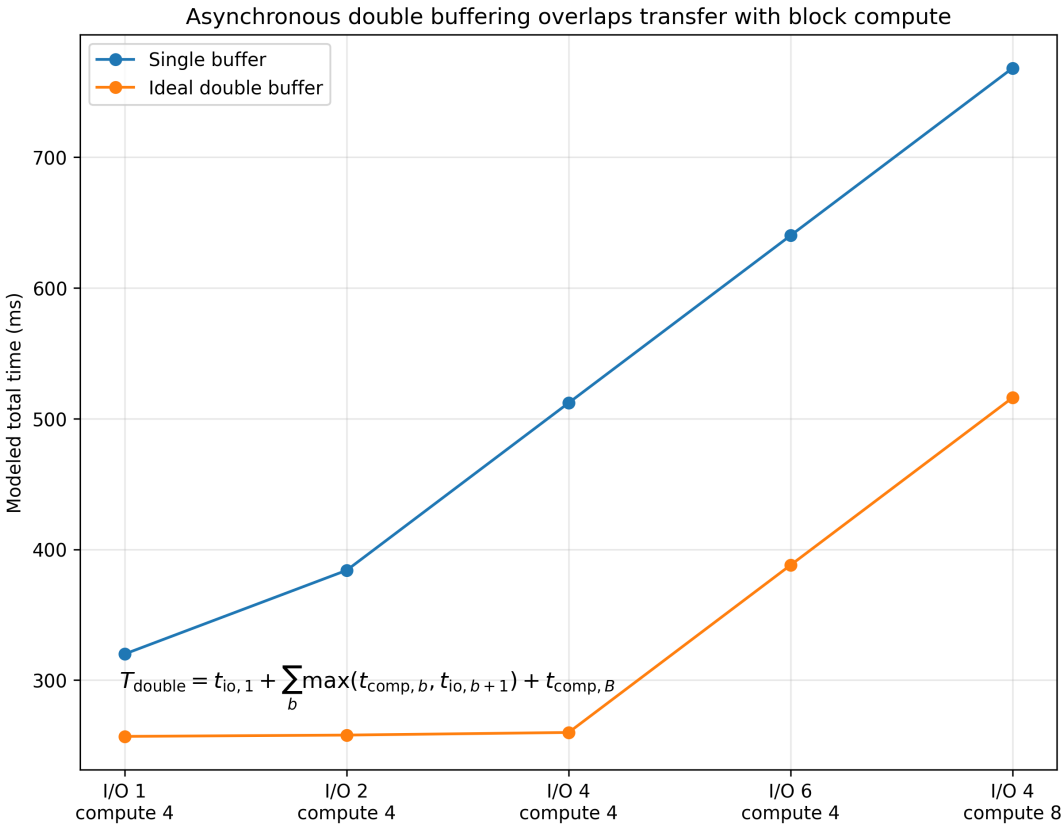


Figure 16: Idealized effect of asynchronous double buffering over 64 streamed blocks. The largest modeled speedup occurs when transfer and computation times are comparable, because neither resource is left idle for most of the pipeline. The additional cost is one extra resident block buffer.

18 Phase-local rank

Let R be a newly generated Krylov residual and

$$R_h = P_h R.$$

Theorem 18.1 (Minimal exact phase width)**.** *Let*

$$r_h = \operatorname{rank}(R_h).$$

Every phase-supported basis spanning $\operatorname{range}(R_h)$ *has at least* r_h *vectors, and a rank-revealing QR or compact SVD attains exactly* r_h*.*

This justifies nonuniform Krylov block widths by phase.

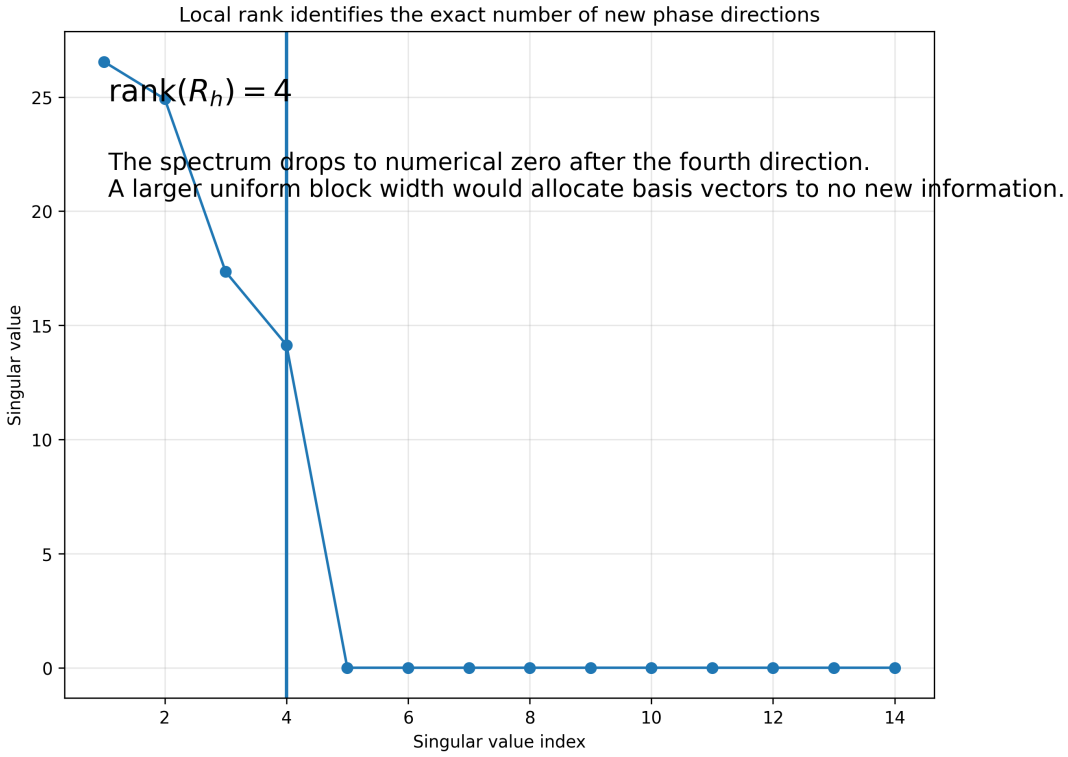


Figure 17: Phase-local rank as a dimension law. The singular values of a phase residual reveal how many genuinely new directions that phase contributes. A rank-revealing factorization keeps exactly those directions instead of allocating a uniform global width.

19 Low-rank interface messages

If an inter-phase coupling has exact factorization

$$P_h H P_g = U_{hg} V_{hg}^\top$$

of rank r_{hg}, then

$$\boxed{P_h H P_g x = U_{hg}(V_{hg}^\top x).}$$

This representation reduces storage and communication. It does not automatically reduce wall-clock time unless the factors are applied through batched or fused kernels; a Python reference implementation can lose to dense BLAS despite lower arithmetic and storage.

20 Exact QR-core updates

Suppose

$$A = QR, \qquad Q^\top Q = I,$$

and append a new block U. Set

$$C = Q^\top U, \qquad F = U - QC, \qquad F = Q_\perp R_\perp.$$

Theorem 20.1 (Exact QR-core update)**.**

$$[A, U] = [Q, Q_\perp] \begin{pmatrix} R & C \\ 0 & R_\perp \end{pmatrix}.$$

Therefore the nonzero eigenvalues of $[A, U][A, U]^\top$ *are exactly the eigenvalues of the small core Gram.*

This is useful for overlapping genomic windows and small incremental variant updates.

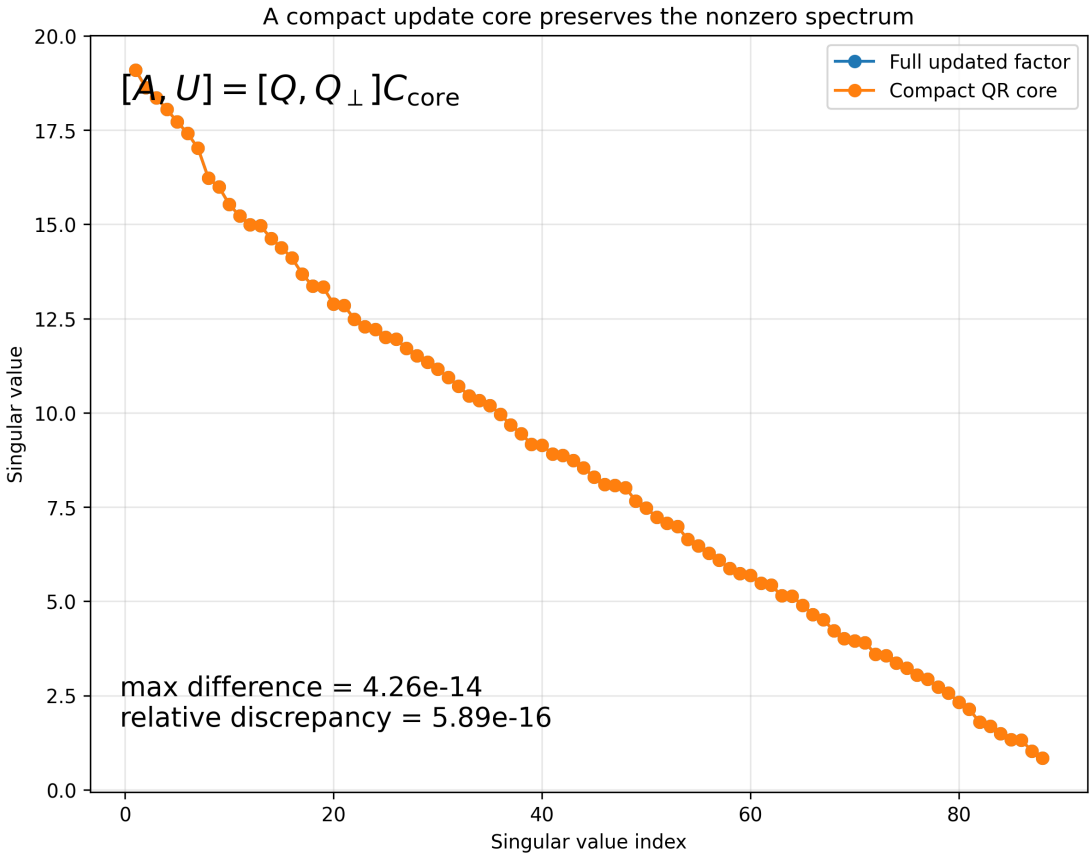


Figure 18: QR-core spectral equivalence. The singular values of the full updated factor and the compact update core lie on top of one another. This permits exact reuse across overlapping windows or appended variant blocks.

21 Range-core spectral compression

Suppose an orthonormal basis Q_r captures the factor range:

$$\text{range}(A) \subseteq \text{range}(Q_r).$$

Then

$$A = Q_r B, \qquad B = Q_r^\top A.$$

Theorem 21.1 (Exact range core)**.** *The nonzero eigenvalues of*

$$K = AA^\top$$

are exactly the eigenvalues of

$$BB^\top.$$

Thus the computational pipeline can terminate its data-streaming stage after the range is captured:

$$\boxed{\text{streamed data} \to \text{small spectral core} \to \text{tail evaluation.}}$$

22 From phase pruning to certified quadratic-form tails

The previous sections certify the effect of phase pruning on Krylov propagation. A SKAT-like test ultimately depends on the spectrum itself, so one more bridge is needed:

$$\text{phase pruning} \longrightarrow \text{operator error} \longrightarrow \text{spectral error} \longrightarrow \text{tail-probability error.}$$

22.1 Block-norm operator domination

Let

$$E = H - H_0$$

be the self-adjoint error introduced by phase pruning. Using the same orthogonal phase projectors P_h, define

$$\mathcal{C}_{hg} = \|P_h E P_g\|_2.$$

Theorem 22.1 (Block-norm operator certificate)**.**

$$\boxed{\|E\|_2 \le \|\mathcal{C}\|_2.}$$

Proof. For any vector x, let

$$r_g = \|P_g x\|_2.$$

Then

$$\|P_h E x\|_2 \le \sum_g \|P_h E P_g\|_2 \, \|P_g x\|_2 = (\mathcal{C} r)_h.$$

Orthogonality of the phase decomposition gives

$$\|Ex\|_2^2 = \sum_h \|P_h E x\|_2^2 \le \|\mathcal{C} r\|_2^2 \le \|\mathcal{C}\|_2^2 \|r\|_2^2 = \|\mathcal{C}\|_2^2 \|x\|_2^2.$$

Taking the supremum over unit vectors proves the claim. □

The important computational point is that $\mathcal{C}$ is only a phase-by-phase matrix. The same block norms already used to construct the polyphase scheduler therefore provide a global operator certificate.

22.2 Weyl enclosure of the quadratic-form coefficients

Assume

$$H \succeq 0$$

and let the eigenvalues of H and H_0, in decreasing order, be

$$\lambda_i \qquad \text{and} \qquad \mu_i.$$

If

$$\|H - H_0\|_2 \le \varepsilon,$$

Weyl's inequality gives

$$|\lambda_i - \mu_i| \le \varepsilon.$$

Corollary 22.2 (Positive spectral interval)**.** *For every i,*

$$\boxed{\ell_i = \max\{0, \mu_i - \varepsilon\} \le \lambda_i \le u_i = \max\{0, \mu_i + \varepsilon\}.}$$

This remains valid even if symmetric phase pruning makes H_0 slightly indefinite.

22.3 Certified generalized-χ^2 tail bracket

For independent standard normal variables Z_i, define

$$Q_- = \sum_i \ell_i Z_i^2, \qquad Q = \sum_i \lambda_i Z_i^2, \qquad Q_+ = \sum_i u_i Z_i^2.$$

Theorem 22.3 (Stochastic tail enclosure)**.** *For every threshold q,*

$$\boxed{\Pr(Q_- \ge q) \le \Pr(Q \ge q) \le \Pr(Q_+ \ge q).}$$

Proof. Couple all three quadratic forms using the same normal variables Z_i. Since

$$0 \le \ell_i \le \lambda_i \le u_i$$

for every i,

$$Q_- \le Q \le Q_+$$

almost surely. Taking upper-tail events yields the result. □

The two endpoint probabilities can be evaluated by the same K-COS machinery used for the unpruned spectrum. Its numerical error certificate is separate from the spectral enclosure above.

22.4 A dual-budget pruning rule

The local Krylov certificate from the previous section and the global spectral certificate answer different questions. The former asks whether omitted couplings matter to the current Krylov query; the latter asks whether they can move the spectrum enough to matter to the final test.

A production schedule should therefore enforce

$$\boxed{\sum_{e \in \mathcal{O}} \beta_e^{(k)} \le \tau_K, \qquad \|\mathcal{C}_{\mathcal{O}}\|_2 \le \tau_\lambda.}$$

The implementation uses a conservative greedy rule: phase pairs are considered in increasing query-risk order, and a pair is omitted only when adding it preserves both inequalities. The rule is certified but is not claimed to be globally optimal under the two simultaneous constraints.

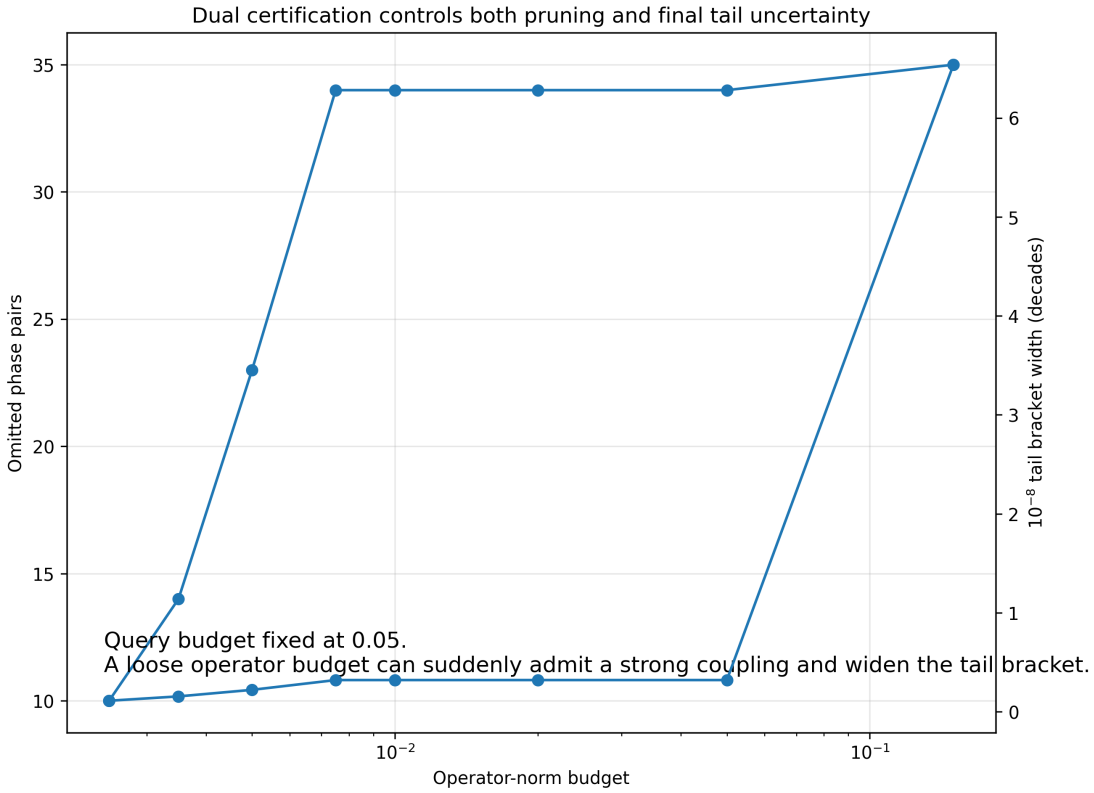


Figure 19: Dual-budget phase pruning. The query-local budget controls Krylov propagation while the operator budget controls final spectral and tail uncertainty. A loose operator budget can admit a strong coupling and widen the tail interval by several decades even when the Krylov-local budget still passes.

In the deterministic ten-phase experiment, fixing the query budget at 0.05 and the operator budget at 0.005 removed 23 of 45 undirected phase pairs. At a true probability of 10^{-8}, the certified tail interval was

$$10^{-8.11143} \le p \le 10^{-7.89036},$$

a width of about 0.221 decades. With an operator budget of 0.0075, 34 pairs were removed and the interval widened to about 0.320 decades. By contrast, allowing a strong coupling to be removed under a very loose operator budget produced a multi-decade interval.

22.5 Certify after prune

There is a second route that does not use the pruned operator as the final spectral approximation. Let U be any orthonormal sample-space basis, possibly generated cheaply from a pruned or approximate Krylov process, and define

$$B = U^\top A, \qquad F = (I - UU^\top)A, \qquad \delta = \|F\|_2.$$

Theorem 22.4 (Range-core singular-value certificate)**.** *Let the singular values of A be $\sigma_i(A)$, and let $\widetilde{\sigma}_i(B)$ denote the singular values of B padded by zeros to the ambient spectral dimension. Then*

$$\boxed{|\sigma_i(A) - \widetilde{\sigma}_i(B)| \le \delta.}$$

Consequently,

$$\max\{0, \widetilde{\sigma}_i(B) - \delta\}^2 \le \lambda_i(AA^\top) \le (\widetilde{\sigma}_i(B) + \delta)^2 .$$

Proof. Because U has orthonormal columns,

$$A = UB + F$$

and

$$\|A - UB\|_2 = \|F\|_2 = \delta.$$

The nonzero singular values of UB are exactly those of B. The standard singular-value perturbation inequality gives the stated bound. Squaring the nonnegative endpoints gives the eigenvalue interval. □

This provides a useful architecture:

pruned operator for basis discovery → one exact streamed factor check → certified spectral core.

22.6 A retained negative result

The preceding theorem is exact, but a correct interval need not be useful.

In a full-rank 50×50 stress test, a pruned Krylov basis of dimensions 12 through 48 left range residuals between roughly 0.87 and 0.99 times the leading singular value. The corresponding 10^{-8} tail intervals were therefore vacuous or thousands of decades wide. At dimension 50, the range residual fell to approximately 1.2×10^{-15}, and the tail interval collapsed to the exact value.

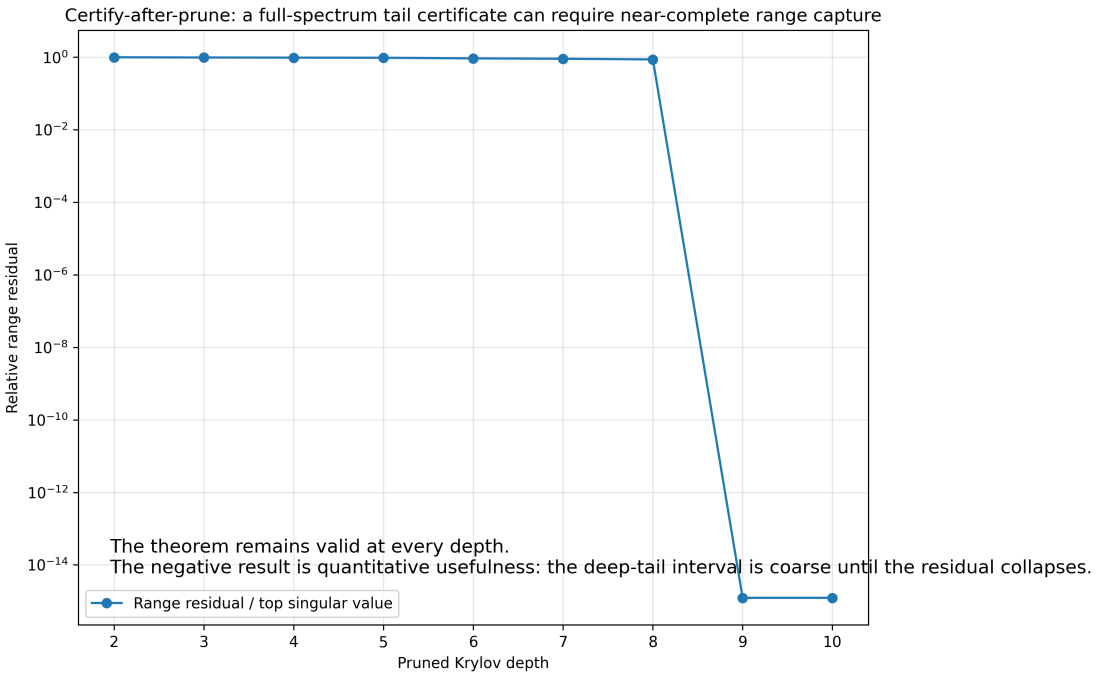


Figure 20: Certify-after-prune range residual. The range-core theorem is valid at every depth, but a deep-tail full-spectrum guarantee can remain useless until the range residual is genuinely small. This negative result is retained as a stopping-rule lesson rather than hidden as an implementation failure.

The practical lesson is that range-core certification is a stopping test, not a promise that a severely incomplete basis automatically yields a useful full-spectrum tail bound.

23 Trace-aware residual certification and exact completion

The spectral bridge in the preceding section used an operator-norm residual as if every unresolved coefficient could move independently by the same amount. That is rigorous but unnecessarily pessimistic once a nontrivial range basis has already been captured. The residual has considerably more structure.

Let $U \in \mathbb{R}^{n\times r}$ have orthonormal columns and define

$$B = U^\top A, \qquad F = (I - UU^\top)A.$$

Then

$$A = UB + F, \qquad U^\top F = 0.$$

Theorem 23.1 (Exact core–residual decomposition)**.**

$$\boxed{A^\top A = B^\top B + F^\top F.}$$

In particular the unresolved spectral term

$$R = F^\top F$$

is positive semidefinite.

Proof. Expand $(UB+F)^\top(UB+F)$. The two cross terms vanish because $U^\top F = 0$, while $U^\top U = I$. □

This changes the certification problem. Instead of an arbitrary Hermitian error, the missing term is a positive residual with measurable spectral moments. Define

$$t = \operatorname{tr} R, \qquad v = \operatorname{tr}(R^2), \qquad \rho = \|R\|_2.$$

Then

$$\boxed{t = \|F\|_F^2, \qquad v = \|FF^\top\|_F^2, \qquad \rho = \|F\|_2^2.}$$

The first quantity is the exact missing mean of the Gaussian quadratic form; the second records how concentrated the residual spectrum is; the third is its largest possible coefficient.

23.1 A trace–Frobenius–cap MGF envelope

Set

$$C = B^\top B, \qquad H = C + R, \qquad c = \lambda_{\max}(C).$$

For $0 < 2\theta(c+\rho) < 1$, put

$$\alpha = (1 - 2\theta c)^{-1}.$$

Theorem 23.2 (Trace–Frobenius–cap log-MGF certificate)**.** *For the Gaussian quadratic form with coefficient matrix H,*

$$K_H(\theta) = -\frac{1}{2}\log\det(I - 2\theta H)$$

satisfies

$$\boxed{K_H(\theta) \le K_C(\theta) + \theta\alpha t + \frac{\theta^2\alpha^2 v}{1 - 2\theta\alpha\rho}.}$$

Proof. Factor the determinant through the known core:

$$I - 2\theta(C + R) = (I - 2\theta C)^{1/2}(I - 2\theta T)(I - 2\theta C)^{1/2},$$

where

$$T = (I - 2\theta C)^{-1/2} R (I - 2\theta C)^{-1/2}.$$

Since

$$\|(I - 2\theta C)^{-1/2}\|_2^2 = \alpha,$$

we have

$$\operatorname{tr} T \le \alpha t, \qquad \operatorname{tr}(T^2) \le \alpha^2 v, \qquad \|T\|_2 \le \alpha\rho.$$

For every eigenvalue x of T,

$$-\frac{1}{2}\log(1 - 2\theta x) \le \theta x + \frac{\theta^2 x^2}{1 - 2\theta\alpha\rho}.$$

The scalar inequality follows from the power series of $-\log(1 - u)$ and a geometric bound on all terms of degree at least two. Summing over the eigenvalues of T proves the result. □

Corollary 23.3 (Trace-aware Chernoff upper tail)**.** *For $Q = Z^\top H Z$ with $Z \sim N(0, I)$,*

$$\boxed{\Pr(Q \ge q) \le \inf_{0<2\theta(c+\rho)<1} \exp\{\widehat{K}_H(\theta) - \theta q\},}$$

where $\widehat{K}_H$ is the preceding MGF upper bound.

A complementary lower endpoint is immediate from

$$H = C + R \succeq C.$$

Using the same normal vector,

$$Z^\top H Z \ge Z^\top C Z$$

almost surely, and therefore

$$\boxed{\Pr(Z^\top C Z \ge q) \le \Pr(Z^\top H Z \ge q).}$$

The lower endpoint can be evaluated by K-COS on the captured core spectrum, while the upper endpoint is obtained from the optimized trace-aware Chernoff bound.

23.2 Finite stress test

The previous coefficient-wise Weyl interval was intentionally conservative. On the same full-rank, 50-dimensional stress test used in the preceding section, its 10^{-8} tail interval remained thousands of decades wide until essentially the complete range had been captured. Preserving the exact residual trace and higher residual moments changes that behavior substantially.

Table 1: Tail-bracket width at a true probability of 10^{-8}. The trace/moment certificate is still conservative, but it is orders of magnitude narrower than coefficient-wise Weyl inflation before full range capture.

Krylov depth	basis dimension	v0.7 Weyl width	v0.8 moment width
4	24	90,097	11.86
5	30	16,511	9.76
6	36	3,747	7.67
7	42	2,275	5.70
8	48	1,165	3.39

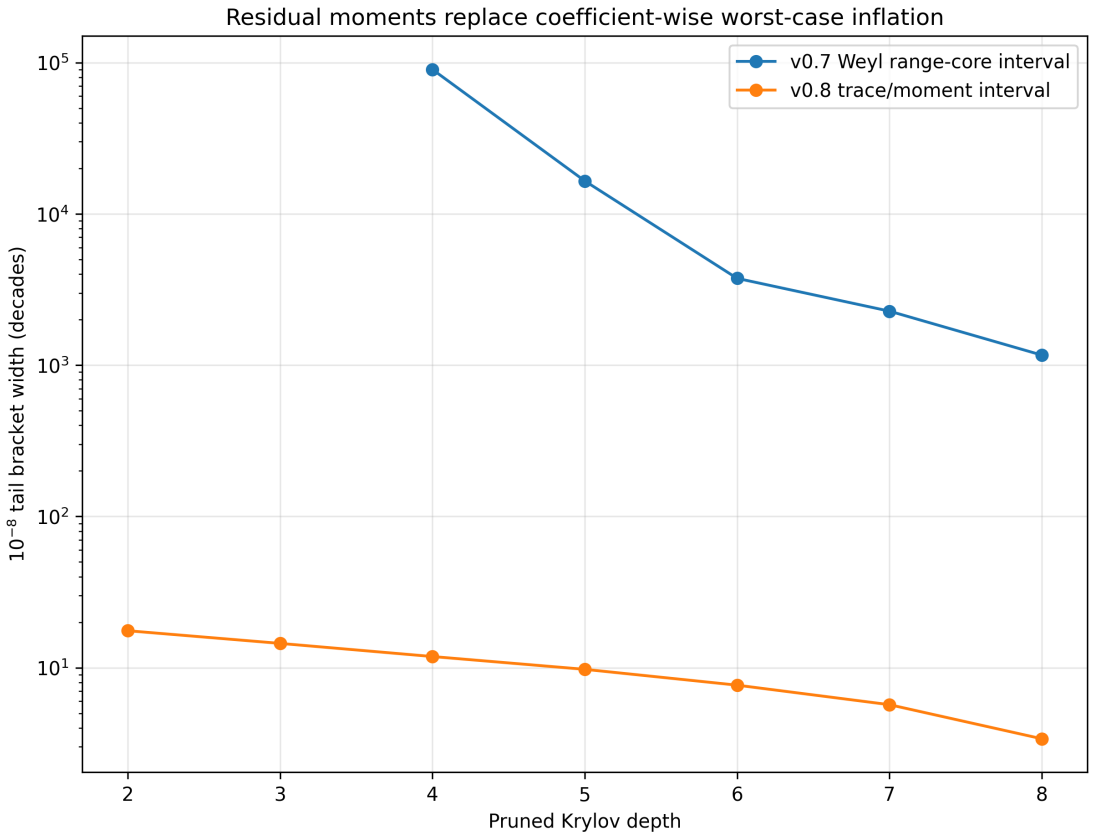


Figure 21: Incomplete-range tail certification before exact completion. The trace/moment bound preserves the amount and concentration of unresolved spectral mass instead of shifting every coefficient by one worst-case radius.

A separate random implementation audit used 120 random factor/range pairs and four admissible positive tilts per pair, for 480 MGF comparisons. Both the trace/cap and trace–Frobenius–cap envelopes dominated the exact log MGF in every tested case. The trace–Frobenius–cap form was at least as tight as the trace/cap form in approximately 83.5% of those comparisons. This finite sweep audits the implementation; the theorem above supplies the universal argument.

23.3 Residual rank from moments

The same invariants determine when it is better to stop approximating the missing spectrum and complete it exactly. Let

$$s = \operatorname{rank} F = \operatorname{rank} R.$$

If the nonzero eigenvalues of R are $r_1, \dots, r_s$, then

$$t = \sum_i r_i, \qquad v = \sum_i r_i^2, \qquad r_i \le \rho.$$

Theorem 23.4 (Residual-rank bracket)**.**

$$\boxed{\left\lceil \max\left\{ \frac{t}{\rho}, \frac{t^2}{v} \right\} \right\rceil \le \operatorname{rank} F \le n - r.}$$

The zero-residual case is interpreted separately as rank zero.

Proof. The inequality $t \le s\rho$ gives $s \ge t/\rho$. Cauchy–Schwarz gives

$$t^2 = \left(\sum_{i=1}^{s} r_i\right)^2 \le s \sum_{i=1}^{s} r_i^2 = sv,$$

so $s \ge t^2/v$. Finally, every column of F lies in $\mathrm{range}(U)^{\perp}$, whose dimension is $n - r$. □

Whenever the integer lower bound equals $n - r$, the missing rank is certified exactly without computing a residual SVD.

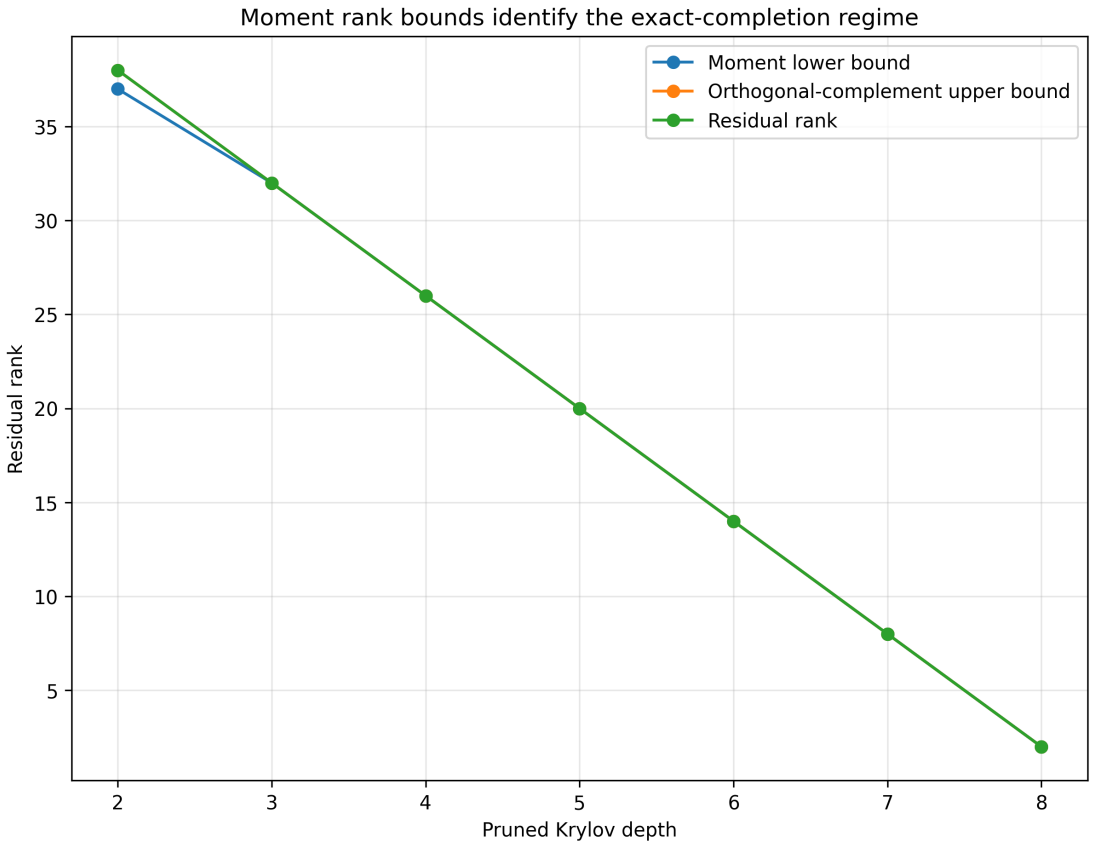


Figure 22: Residual-rank certification in the full-rank stress test. From depth 3 onward the moment lower bound meets the orthogonal-complement upper bound. At depth 8 exactly two residual directions remain.

23.4 Exact residual completion

Suppose a compact factorization of the residual is

$$F = WD,$$

where W has orthonormal columns. Since F lies in the orthogonal complement of U,

$$U^{\top} W = 0.$$

Theorem 23.5 (Exact residual completion)**.**

$$\boxed{A = [U\ W] \begin{bmatrix} B \\ D \end{bmatrix}.}$$

Consequently, the nonzero singular values of A are exactly the singular values of the stacked completion core

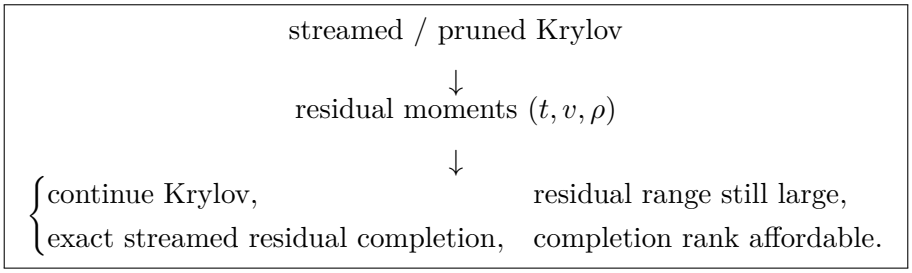
$$\mathcal{C}_{\text{complete}} = \begin{bmatrix} B \\ D \end{bmatrix}.$$

This identity admits a column-streamed implementation. For each residual column block, project against the residual basis accumulated so far, append only genuinely new directions through rank-revealing QR or SVD, and extend the coordinate matrices of earlier blocks with zero rows. New residual directions are orthogonal to every earlier residual block by construction, so no earlier data block must be revisited.

In the deterministic stress test, exact residual completion reproduced the full singular spectrum at relative errors between approximately 2×10^{-16} and 6×10^{-16} for all tested incomplete bases. The resulting 10^{-8} tail differed from the full-spectrum reference by less than 10^{-14} decades. At depth 8 the candidate range had dimension 48, the missing rank was certified as exactly two, and appending those two residual directions closed the spectrum without another approximate Krylov stage.

23.5 Hybrid stopping rule

The resulting production logic is not "approximate forever." It is a certified handoff between two regimes:



The trace-aware tail certificate remains useful as a rigorous intermediate bound and as a stopping diagnostic. Once only a small number of residual directions remain, exact completion is mathematically preferable to forcing a worst-case tail envelope to become artificially sharp.

24 Rare-variant kernel case study

The preceding sections establish operator identities and certified numerical bounds. This section tests the same machinery on the project's existing rare-variant SKAT simulation layer, including explicit LD structure, covariates, per-variant standardization, rare-variant weighting, and the phenotype-space score statistic.

24.1 Simulation provenance and implementation correction

Two established genotype generators were used.

The first is a latent-Gaussian block-LD generator. Within a genomic block, alleles are obtained by thresholding correlated Gaussian latent variables at variant-specific minor-allele frequencies.

The second is a copy-LD generator in which haplotype states persist along a block with high probability and are occasionally redrawn. During the v0.9 audit, a reproducibility defect was found in an earlier simulator implementation: the genotype matrix was allocated with `np.empty` and then updated using

`+=`. Because the initial contents of `np.empty` are unspecified, those old copy-LD benchmark values are not reproducible evidence. The v0.9 generator uses an explicitly zero-initialized matrix before haplotype accumulation. All copy-LD results reported below use the corrected implementation.

24.2 Tested LD regimes

Four deterministic scenarios were evaluated.

Table 2: Realistic rare-variant simulation scenarios. Adjacent r^2 is computed within the simulated LD blocks after variant standardization.

scenario	n	m	rank	median adjacent r^2	effective rank
latent LD, moderate	600	3000	596	0.0069	441.9
latent LD, strong	600	3000	596	0.1140	240.3
copy LD, strong	600	3000	596	0.8298	200.1
copy LD, large	1000	5000	996	0.8179	353.0

The corresponding spectra therefore span a substantial range of concentration: stronger LD reduces effective rank even when the nominal quadratic-form rank remains close to the sample residual dimension.

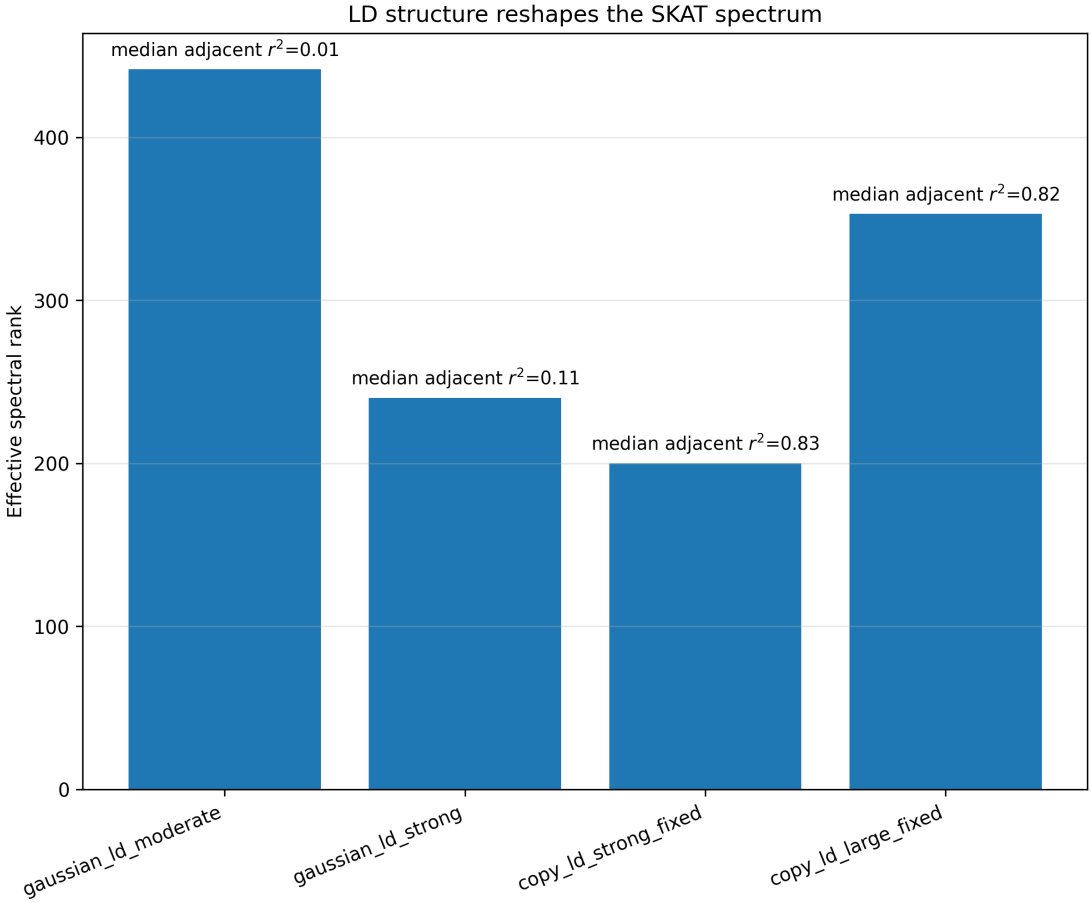


Figure 23: LD structure changes the effective spectral rank. The corrected copy-LD models create much stronger local dependence than the latent-Gaussian moderate-LD case, while all remain within the same streamed SKAT pipeline.

24.3 Streaming directly from raw genotype blocks

The raw genotype matrix was streamed in resident variant blocks of widths 31, 127, and 509. Within each block the implementation performs

$$G_b \to G_b - \bar{G}_b \to \text{variance scaling} \to M_X G_b \to W_b^{1/2} \to A_b A_b^\top,$$

without materializing the full weighted factor.

Across all four LD scenarios and all three resident block sizes, the largest relative discrepancy against the fully materialized kernel was

$$\boxed{6.61 \times 10^{-16}}.$$

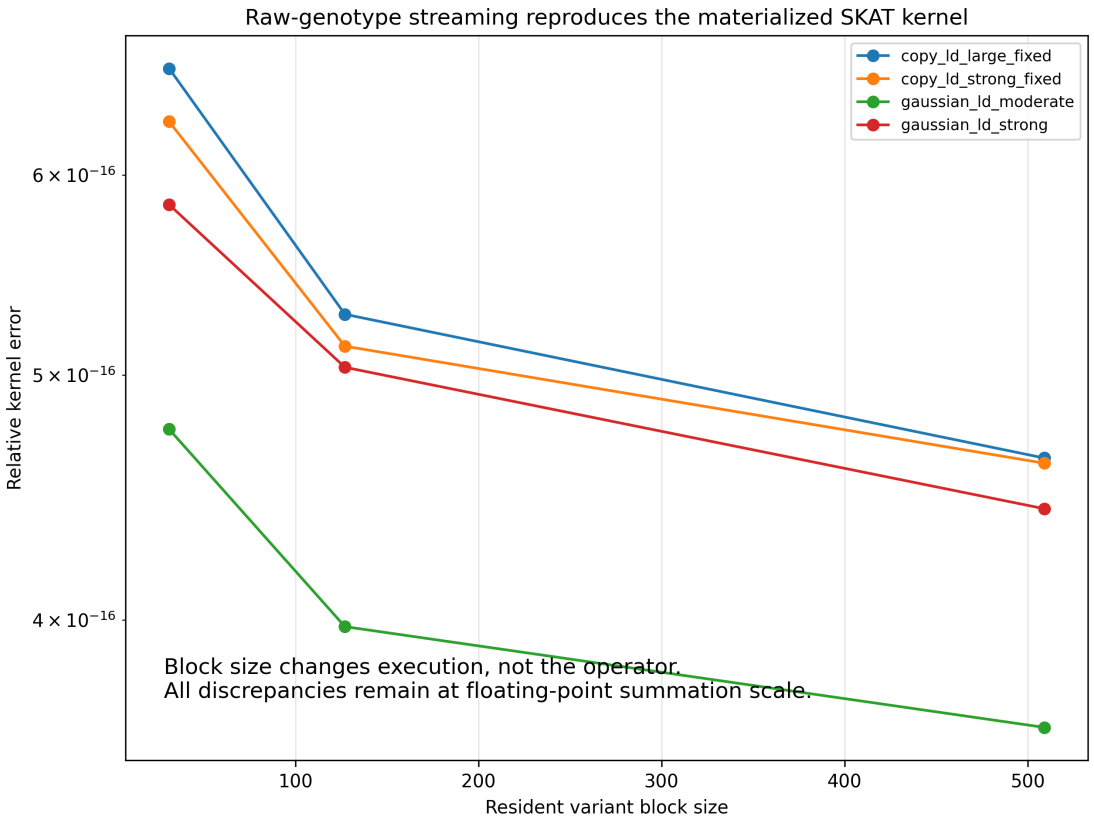


Figure 24: Raw-genotype streaming reproduces the materialized SKAT kernel across multiple LD regimes and block sizes. The observed discrepancies are at floating-point summation scale.

This test is stronger than applying the streamed identity to a precomputed factor because the centering, variance scaling, covariate projection, and rare-variant weights are all performed inside the block loop.

24.4 Phenotype-space equivalence

For each scenario, 100 Gaussian null residual draws were evaluated in three ways:

1. directly through the factor,
$$Q = \frac{1}{2}\|A^\top e\|_2^2;$$
2. through the sample kernel,
$$Q = \frac{1}{2} e^\top K e;$$
3. through the eigencoordinates of K.

The worst relative discrepancy across all draws and all scenarios was

$$\boxed{8.32 \times 10^{-16}}.$$

Thus the simulated phenotype path, the streamed kernel, and the generalized χ^2 spectral representation agree to floating-point precision.

24.5 Deep tails on realistic LD spectra

For every scenario, tail queries were generated at tilt fractions 0.90, 0.97, and 0.99 of the positive singularity. K-COS was compared with a higher-resolution full-spectrum tilted reference.

The large corrected copy-LD case was then pushed further, to fractions 0.995, 0.997, 0.999, and 0.9995. The deepest tested reference value was

$$\boxed{\log_{10} p = -457.973916}.$$

At that point the displayed K-COS and high-resolution full-spectrum values were identical. Across the complete realistic-LD battery, the largest absolute discrepancy was

$$\boxed{3.21 \times 10^{-12} \text{ decades}}.$$

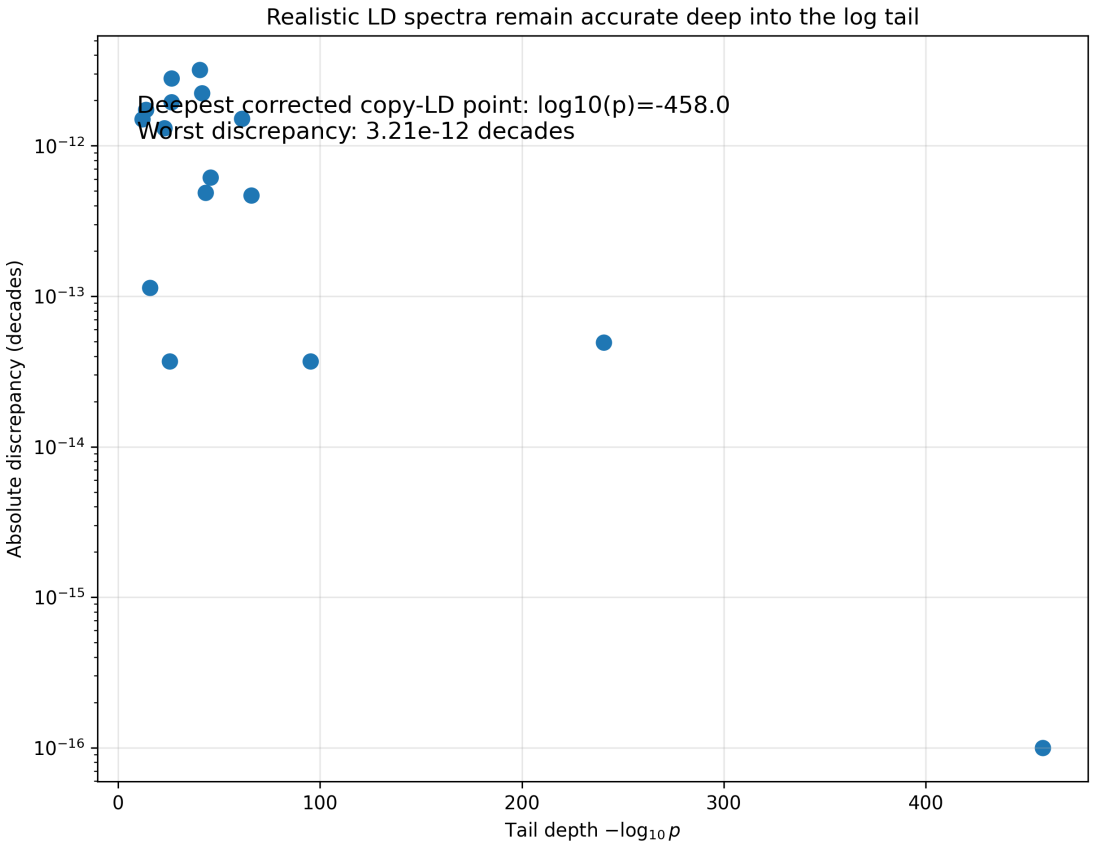


Figure 25: K-COS versus the high-resolution full-spectrum tilted reference on realistic LD spectra. Accuracy remains at roughly machine-scale in log probability through the deepest corrected copy-LD query, near 10^{-458}.

For the $n = 1000$, $m = 5000$ corrected copy-LD spectrum, selected results are

tilt fraction	$\log_{10} p$	K-COS K	absolute error (decades)
0.9950	-65.950669	2048	4.69×10^{-13}
0.9970	-95.305034	2048	3.70×10^{-14}
0.9990	-240.602047	8192	4.94×10^{-14}
0.9995	-457.973916	16384	below stored reference resolution

24.6 Interpretation and boundary

This simulation block does not add a new algebraic theorem. It supplies a more demanding validation layer for the theorem chain already proved. Specifically, it shows that the exact streaming identities and the K-COS deep-tail machinery remain numerically stable under substantially different rare-variant LD structures and through the complete raw-genotype preprocessing path.

The evidence remains synthetic. These generators do not reproduce every population-genetic feature of a sequencing cohort, and this block is not an external-cohort replication. It also addresses the Gaussian linear-model null; finite-sample Bernoulli or logistic calibration remains a separate statistical problem.

25 General production API and deep-tail arithmetic hardening

The preceding mathematics is exposed through a domain-neutral engine rather than through a SKAT-only entry point. The central positive engine accepts:

- an explicit nonnegative spectrum;
- a positive semidefinite matrix;
- a dense factor A;
- a replayable streamed column-factor source;
- or a covariance/quadratic-matrix pair (Σ, M).

A signed central engine accepts a signed spectrum, a symmetric operator, or

$$(\Sigma, M), \qquad M = M^\top,$$

and performs the exact central-indefinite reduction proved in Section 1. A separate noncentral positive engine accepts

$$(\mu, \Sigma, M), \qquad M \succeq 0,$$

and performs the corresponding noncentral reduction.

25.1 Log-domain packet error accumulation

At extreme tail depths the analytic packet certificate contains sums of the form

$$\frac{2}{U} \sum_k |I_k| \left(e^{\varepsilon_k} - 1\right).$$

Direct evaluation can overflow even when the candidate is destined to be rejected by the adaptive planner. The hardened implementation instead computes

$$\log\left(\frac{2}{U} \sum_k |I_k| \left(e^{\varepsilon_k} - 1\right)\right)$$

by log-sum-exp arithmetic, and compares this quantity with the logarithm of the requested tolerance before converting a final accepted bound back to float64.

This change does not weaken the mathematical bound; it changes only its floating-point representation.

25.2 Reference convergence is part of verification

An independently evaluated reference is useful only if its own numerical resolution has converged. In a rank-350 deep-tail audit, an initial fixed-grid reference appeared to disagree with the adaptive answer. Increasing the reference COS grid gave discrepancies of approximately

$$1.88 \times 10^{-2}, \quad 2.6 \times 10^{-3}\text{–}4.0 \times 10^{-3}, \quad 1.39 \times 10^{-4}, \quad 2.60 \times 10^{-8}$$

decades as the grid was refined through

$$K = 4096,\ 8192,\ 16384,\ 32768.$$

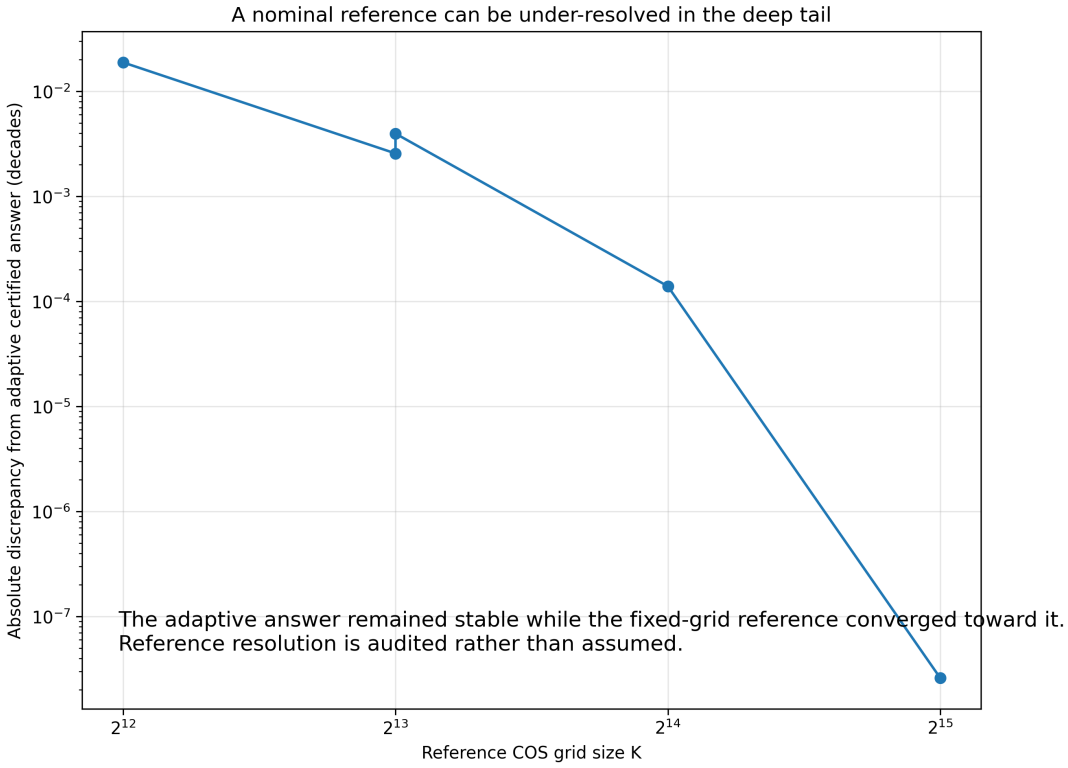


Figure 26: Reference-convergence audit. The adaptive certified answer remains stable while an initially under-resolved fixed-grid reference converges toward it. A single nominally large reference grid is therefore not treated as authoritative without its own refinement check.

The production verification rule is consequently two-sided: the adaptive solver must satisfy its own certificate, and any external or fixed-grid reference used for comparison must demonstrate numerical convergence.

26 Interface to spectral packets and K-COS

The matrix packets above operate before eigendecomposition. They should be distinguished from the spectral packets used after the eigenvalues λ_j are known.

The complete stack is

$$
\begin{aligned}
&G \\
&\quad \to \text{storage blocks} \\
&\quad \to \text{matrix phases} \\
&\quad \to \text{streamed } A, A^\top, K \\
&\quad \to \text{polyphase/rank-adaptive Krylov} \\
&\quad \to \text{small spectral core} \\
&\quad \to \{\lambda_j\} \\
&\quad \to \text{centered spectral packets} \\
&\quad \to \text{deep-tail K-COS.}
\end{aligned}
$$

Storage decomposition, structural decomposition, spectral compression, and tail compression are independent axes and can be active simultaneously.

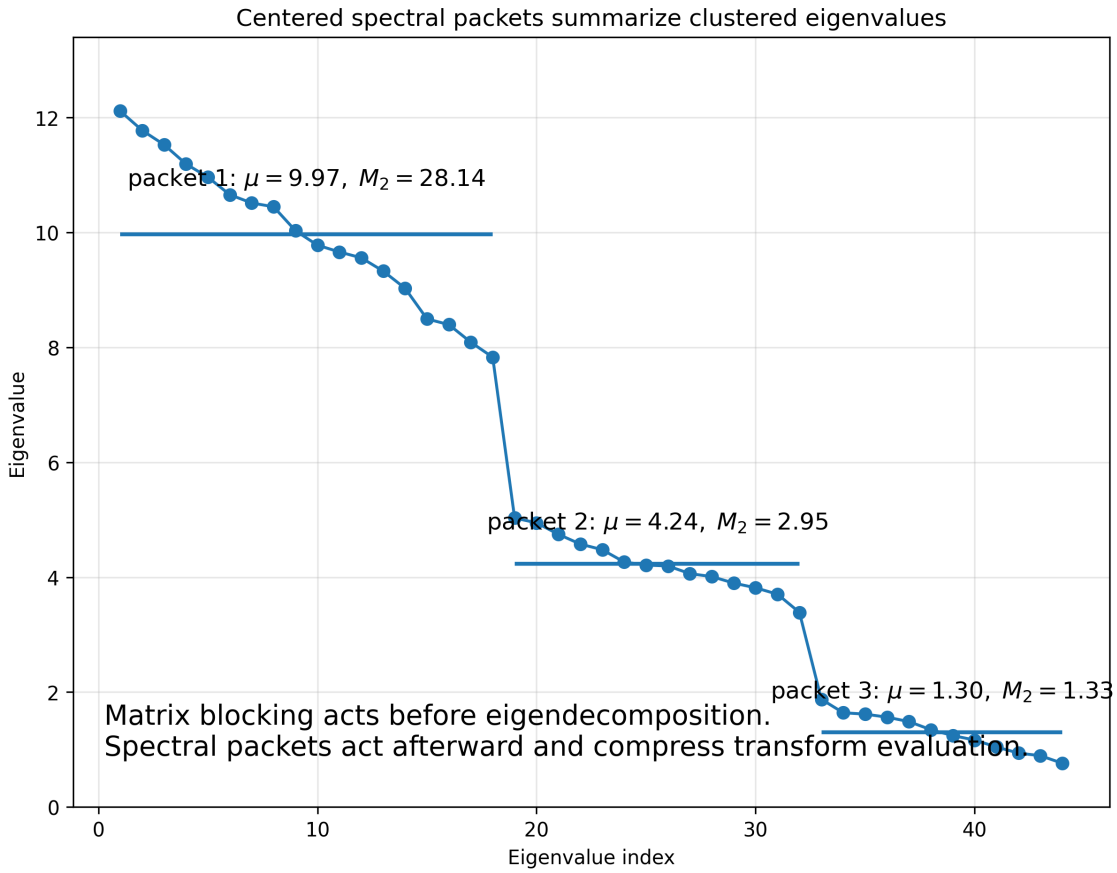


Figure 27: Spectral packets form a second, distinct decomposition layer. After the large matrix problem has been reduced to eigenvalues, nearby eigenvalues are represented by packet centers and centered moments for K-COS transform evaluation. These spectral packets should not be confused with the storage blocks used before eigendecomposition.

27 Memory planning law

Let d be the number of bytes per floating scalar, c the resident variant count, s the Krylov width, and p the covariate rank. Ignoring library workspace and allocator overhead, a streamed sample-space action has working memory on the order of

$$\boxed{d\,[nc + 2ns + cs + np + O(ps + c)].}$$

The full factor costs

$$dnm.$$

For fixed n, p, s, c,

$$\boxed{\frac{M_{\text{stream}}}{M_{\text{full}}} = O\left(\frac{c}{m} + \frac{s}{m} + \frac{p}{m}\right),}$$

so device memory remains bounded as total variant count grows.

Table 3: Algebraic float64 memory model for $n = 2000$, $p = 4$, Krylov width $s = 16$, resident variant block $c = 1024$.

m	full factor (MiB)	streamed KV (MiB)	tiled Gram (MiB)
10,000	152.6	16.308	8.555
50,000	762.9	16.308	8.555
100,000	1,525.9	16.308	8.555
500,000	7,629.4	16.308	8.555
1,000,000	15,258.8	16.308	8.555

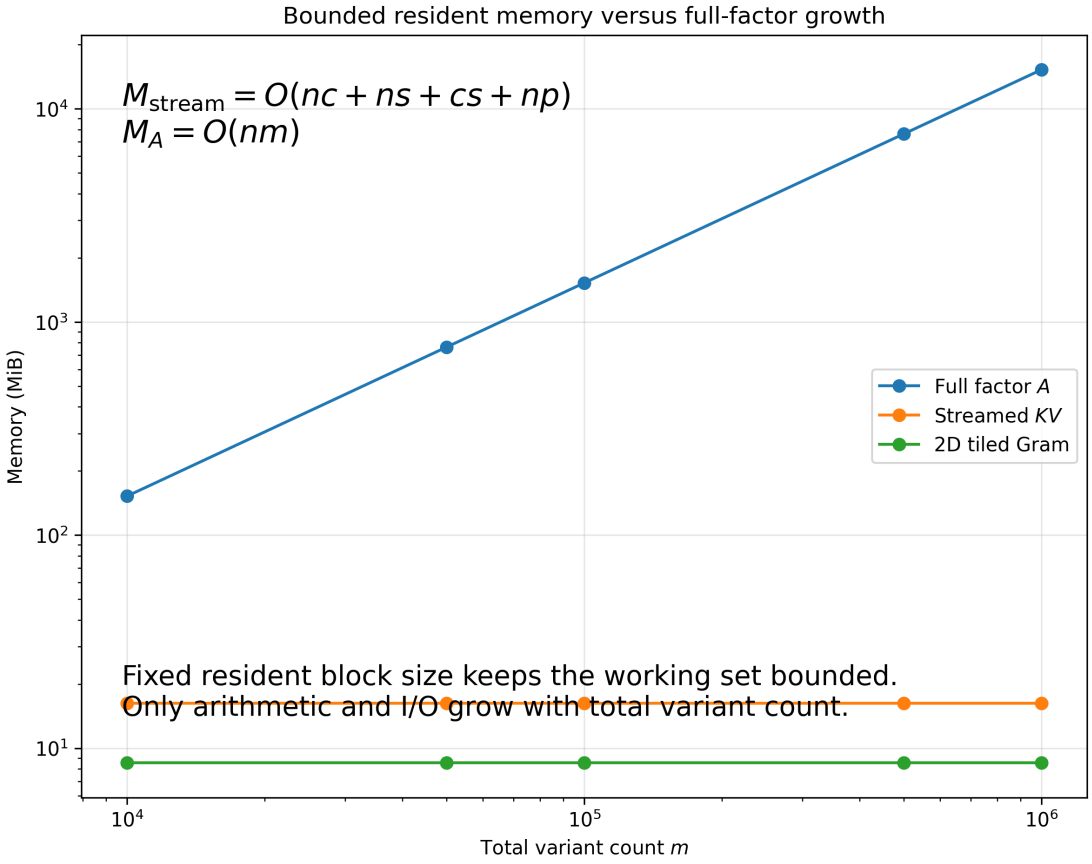


Figure 28: The memory law. The complete factor grows linearly with total variant count m, whereas the streamed kernel action and the two-dimensional tiled construction remain bounded when resident block size, Krylov width, and covariate rank are fixed.

28 Numerical verification

All identities were compared against explicitly formed reference matrices.

Table 4: Partition-invariance checks.

partition	blocks	Gram relative error	action relative error
coarse 4	4	2.91×10^{-16}	5.99×10^{-16}
fine 64	32	3.39×10^{-16}	6.00×10^{-16}
irregular	9	3.34×10^{-16}	5.87×10^{-16}
singleton-front	21	2.43×10^{-16}	5.82×10^{-16}
permuted, block 73	29	3.52×10^{-16}	5.89×10^{-16}

A hierarchical coarse-to-fine refinement gave relative discrepancy

$$3.17 \times 10^{-16}.$$

For an explicitly PSD polyphase operator

$$H = L^{\top} L$$

with block-bidiagonal L, the smallest eigenvalue was approximately

$$2.86,$$

and the largest norm observed outside the exact k-step light cone was

$$\boxed{0}.$$

The QR-core update reproduced the full singular spectrum with relative ℓ_2 discrepancy

$$5.89 \times 10^{-16}.$$

29 Method synthesis: commuting decompositions

The solver is organized around four independent decomposition axes:

$$\boxed{\mathcal{D} = \mathcal{D}_{\text{storage}} \circ \mathcal{D}_{\text{phase}} \circ \mathcal{D}_{\text{rank}} \circ \mathcal{D}_{\text{spectrum}}.}$$

They answer different questions. Storage decomposition controls what is resident. Phase decomposition controls where information can propagate. Rank decomposition controls how many directions must be retained. Spectral packetization controls the cost of the final deep-tail transform. The central structural fact is that these representations can be changed independently without changing the exact operator when their respective invariance conditions are satisfied.

29.1 Storage-phase commutation

Fix a structural phase partition

$$\{1, \dots, m\} = \mathcal{P}_1 \sqcup \cdots \sqcup \mathcal{P}_H$$

and let each phase be subdivided into arbitrary storage blocks

$$\mathcal{P}_h = J_{h1} \sqcup \cdots \sqcup J_{hB_h}.$$

Let S denote this physical storage subdivision. The phase projectors P_h are defined by the structural phases, not by S.

Theorem 29.1 (Storage-phase commutation theorem)**.** *Let*

$$A = M_X G W^{1/2}, \qquad H = A^\top A,$$

and let S and S' be any two exact storage refinements that respect the same structural phase partition. Then:

(i) the exact operator H is identical under S and S';

(ii) every phase coupling $P_h H P_g$ is identical under S and S';

(iii) the exact phase graph and every k-step polyphase light cone are identical under S and S';

(iv) any exact matrix-free Krylov method applied through streamed products produces the same Krylov subspaces in exact arithmetic under S and S'.

Proof. Storage refinement changes only the order and grouping of finite sums. By partition refinement invariance, the streamed products reconstruct the same A, $A^\top$, and therefore the same $H = A^\top A$. Since each P_h is fixed by the structural phase partition, every block $P_h H P_g$ is unchanged. The phase graph is defined only by these blocks, so its path structure and light cones are unchanged. Finally, an exact matrix-free Krylov method depends only on repeated applications of the same operator to the same starting block; hence its generated subspaces are identical. □

Corollary 29.2 (Adaptive resident-block scheduler)**.** *An implementation may change resident block size at runtime—including recursively splitting a phase block after an allocation failure—without changing the exact phase graph or exact Krylov trajectory, provided all required storage blocks are processed exactly.*

This separates *mathematical locality* from *physical storage granularity*. A phase may remain fixed as an LD window or annotation class while its physical representation is subdivided according to free GPU memory, scalar precision, transfer granularity, or stream count.

29.2 Frontier-to-I/O theorem

Theorem 29.3 (Exact frontier-to-I/O reduction)**.** *Suppose the exact phase graph of H is used and the current Krylov block is supported on a phase set F_0. At depth k, let F_k be the graph k-neighborhood of F_0. Then no storage block lying wholly in a phase outside F_k can affect the k-step Krylov vector. Such a block need not be loaded during that step.*

Proof. For every phase $h \notin F_k$ and every starting phase $g \in F_0$, the polyphase light-cone theorem gives $P_h H^k P_g = 0$. Storage-phase commutation allows each active phase operator to be evaluated through arbitrary exact storage blocks. Therefore blocks outside the active structural frontier contribute identically zero to the requested iterate and may be omitted. □

The theorem turns an algebraic support statement into an execution law:

$$\boxed{\text{Krylov depth} \longrightarrow \text{reachable phases} \longrightarrow \text{storage blocks that must be transferred.}}$$

This is stronger than ordinary chunking, because the scheduler is constrained by the exact operator graph rather than solely by available memory.

29.3 Full method stack

The complete positive-form solver may therefore be written as

$$\begin{aligned} A &\longrightarrow \text{resident storage blocks} \\ &\longrightarrow \text{structural phase frontier} \\ &\longrightarrow \text{phase-local Krylov/range core} \\ &\longrightarrow \text{residual moment certificate or exact completion} \\ &\longrightarrow \{\lambda_j\} \\ &\longrightarrow \text{centered spectral packets} \\ &\longrightarrow \log P(Q \geq q). \end{aligned}$$

Three invariances make the stack composable:

1. storage refinement does not change the represented operator;
2. structural phase scheduling does not depend on physical block size;
3. once the spectrum or a certified spectral enclosure is obtained, the tail engine is independent of the application that produced it.

The resulting method can be aggressive during basis discovery and still return to exactness. Approximate phase pruning is controlled by query and operator budgets; unresolved range structure is measured by trace, square-trace, and spectral-cap invariants; and when the remaining rank is small, streamed residual completion restores the exact nonzero spectrum before the final tail evaluation.

29.4 Method summary

The resolved synthesis is

$$\begin{aligned} &\text{exact streamed factor actions} + \text{ structural light cones} + \text{ certified pruning} \\ &+ \text{ phase-local rank} + \text{ residual moment certificates} + \text{ exact residual completion} \\ &+ \text{ spectral packets} + \text{ log-domain extreme-tail evaluation.} \end{aligned}$$

The central design principle is separation of representation from semantics: changing how an operator is stored, scheduled, compressed, or evaluated must not silently change the mathematical quadratic form. Each approximation layer therefore either carries an explicit certificate or terminates in an exact completion step.

30 Guarantee levels and scope

The framework contains several different evidence levels. They are kept separate because a proved algebraic identity, a finite numerical validation, and a directed-rounding probability enclosure are different kinds of claims.

Statement	Status	Meaning
Block factor and streamed-action identities	proved	Exact finite-dimensional matrix identities; storage blocking introduces no mathematical approximation.
Storage-phase commutation and exact light cones	proved	Physical block refinement does not change the structural phase graph; phases outside the exact finite-depth frontier contribute zero.
Approximate light-cone and pruning certificates	proved + tested	Weak discarded couplings carry explicit finite-depth transfer and operator-norm budgets.
Range residual and exact completion	proved + tested	The unresolved factor is measured directly; small residual rank can be completed exactly.
Positive exact-spectrum tail enclosure	directed rounding + proved remainders	For rank > 2, the finite transform is enclosed numerically and enlarged by proved support-/alias and infinite-mode bounds.
Accuracy-driven positive tail solver	certified / fail-closed	A result is accepted only after $p_+/p_- - 1 \leq \varepsilon_{\text{requested}}$ is proved.
Positive noncentral engine	exact reduction + tested tail solver	The Gaussian form is reduced to a deterministic shift plus positive noncentral chi-square components.
Signed / indefinite engine	proved reduction + mixed numerical certification	High-rank and finite-endpoint cases are certified; low-rank mixed-sign forms remain limited by slow Fourier decay of the present COS certificate.
Binary64 factor to rigorous spectrum	implemented at moderate dimension	Candidate eigensystems are post-certified by directed-rounding orthogonality and reconstruction bounds, then propagated monotonically into the tail. The remaining issue is scalable streamed/subspace verification rather than the basic correctness theorem.

Three boundaries remain material:

1. scale the verified matrix-to-spectrum bridge to streamed range-core and residual-completion linear algebra without materializing a full dense basis;
2. replace the conservative low-rank mixed-sign Fourier-tail certificate with a rigorous convolution or quadrature route;
3. implement the proven streamed and residual-completion algorithms in the final native GPU runtime and measure transfer/compute crossover on hardware.

These are boundaries of the implementation, not hidden assumptions in the certified positive matrix-to-tail result at moderate dimension.

31 Conclusion

The quadratic-form problem can be separated into representation, structure, spectral recovery, and tail evaluation.

At the representation level, factors can be streamed in blocks without changing the operator. At the structural level, phase graphs constrain finite-depth propagation and allow certified scheduling. At the spectral level, unresolved range structure is measured rather than guessed and can be completed

exactly when its rank becomes small. At the probability level, exponential tilting and spectral packets make extreme tails computationally tractable, while directed-rounding interval arithmetic and analytic remainder bounds turn the positive exact-spectrum calculation into an explicit probability enclosure.

The resulting numerical contract is deliberately stronger than "approximately evaluate a small probability." For the rigorous positive backend, the user supplies an accuracy requirement. Internal resolution is refined until that requirement is proved, or the solver returns an uncertified interval rather than claiming success.

The deepest exact control demonstrated here reaches

$$p = 10^{-5000}.$$

At that point the accuracy-driven backend can certify relative multiplicative uncertainty below 10^{-18}. This establishes that extreme-tail depth by itself is not the limiting numerical variable; the meaningful limit is whether the complete error ledger can be made smaller than the requested tolerance.

The matrix-to-spectrum bridge now closes that upstream correctness gap for moderate binary64 factors by post-certifying an otherwise untrusted candidate eigensystem. The remaining rigorous frontier is primarily one of scalable representation: carry the same enclosure through streamed range cores and residual completion without forming a full dense eigenbasis, while extending the fail-closed accuracy contract to difficult low-rank mixed-sign forms.

A Reference pseudocode

```
def stream_KV(block_source, V, project=lambda x: x):
    U = project(V)
    Y = zeros_like(U)
    for A_block in block_source:
        Y += A_block @ (A_block.T @ U)
    return project(Y)

def certified_tail(q, spectrum, relative_accuracy):
    # Refines support, COS modes, and interval precision.
    # Returns certified=False if the requested accuracy is not proved.
    return certified_positive_logsf(
        q, spectrum, relative_accuracy=relative_accuracy
    )
```

B Reproducibility and evidence package

The release accompanying this manuscript contains the executable engine, interval backend, deterministic regression tests, numerical evidence tables, and proof-audit notes used to support the quantitative claims.

The principal production-facing files are:

- `quadratic_form_engine.py`: general central/noncentral interface;
- `accuracy_driven.py`: fail-closed tolerance-driven controller;
- `interval_kcos.py`: directed-rounding positive-tail backend;

- `kcos_v2_hardened.py`: log-domain adaptive positive-tail engine;
- `generalized_tail.py`: signed and noncentral generalized forms;
- `residual_completion.py`: exact completion of unresolved factor range.

Evidence tables record exact-control, realistic-spectrum, signed-form, noncentral, interval-enclosure, and requested-accuracy experiments. The release also records negative results and corrected implementation defects so that failed reference calculations or superseded simulation defects are not silently reused as evidence.